Cancer Statistics, 2008

Ahmedin Jemal, DVM, PhD; Rebecca Siegel, MPH; Elizabeth Ward, PhD; Yongping Hao, PhD; Jiaquan Xu, MD; Taylor Murray; Michael J. Thun, MD, MS*

ABSTRACT Each year, the American Cancer Society estimates the number of new cancer cases and deaths expected in the United States in the current year and compiles the most recent data on cancer incidence, mortality, and survival based on incidence data from the National Cancer Institute, Centers for Disease Control and Prevention, and the North American Association of Central Cancer Registries and mortality data from the National Center for Health Statistics. Incidence and death rates are age-standardized to the 2000 US standard million population. A total of 1,437,180 new cancer cases and 565,650 deaths from cancer are projected to occur in the United States in 2008. Notable trends in cancer incidence and mortality include stabilization of incidence rates for all cancer sites combined in men from 1995 through 2004 and in women from 1999 through 2004 and a continued decrease in the cancer death rate since 1990 in men and since 1991 in women. Overall cancer death rates in 2004 compared with 1990 in men and 1991 in women decreased by 18.4% and 10.5%, respectively, resulting in the avoidance of over a half million deaths from cancer during this time interval. This report also examines cancer incidence, mortality, and survival by site, sex, race/ethnicity, education, geographic area, and calendar year, as well as the proportionate contribution of selected sites to the overall trends. Although much progress has been made in reducing mortality rates, stabilizing incidence rates, and improving survival, cancer still accounts for more deaths than heart disease in persons under age 85 years. Further progress can be accelerated by supporting new discoveries and by applying existing cancer control knowledge across all segments of the population. (*CA Cancer J Clin 2008;58:71–96.*) © *American Cancer Society, Inc., 2008.*

Dr. Jemal is Strategic Director, Cancer Surveillance, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Ms. Siegel is Manager, Surveillance Information Services, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Dr. Ward is Managing Director, Surveillance Research, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Dr. Hao is Senior Epidemiologist, Surveillance Research, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Dr. Xu is Epidemiologist, Mortality Statistics Branch, Division of Vital Statistics, National Center for Health Statistics, Centers for Disease Control and Prevention, Hyattsville, MD.

Mr. Murray is Manager, Surveillance Data Systems, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Dr. Thun is Vice President, Department of Epidemiology and Surveillance Research, American Cancer Society, Atlanta, GA.

Published online through *CA First Look* at http://CAonline.AmCancerSoc.org.

DOI: 10.3322/CA.2007.0010

INTRODUCTION

Cancer is a major public health problem in the United States and many other parts of the world. Currently, one in 4 deaths in the United States is due to cancer. In this article, we provide an overview of cancer statistics, including updated incidence, mortality, and survival rates, and expected numbers of new cancer cases and deaths in 2008.

MATERIALS AND METHODS

Data Sources

Mortality data from 1930 to 2005 in the United States were obtained from the National Center for Health Statistics (NCHS).[1] Incidence data for long-term trends (1975 to 2004), 5-year relative survival rates, and data on lifetime probability of developing cancer were obtained from the Surveillance, Epidemiology, and End Results (SEER) program of the National Cancer Institute, currently covering about 26% of the US population.[2–5] Incidence data (1995 to 2004) for projecting new cancer cases were obtained from cancer registries that participate in the SEER program or the Centers for Disease Control and Prevention (CDC)'s National Program of Cancer Registries (NPCR) through the North American Association of Central Cancer Registries (NAACCR). State-specific incidence rates were

*Disclaimer: The findings and conclusions in this report are those of the author and do not necessarily represent the views of the Centers for Disease Control and Prevention.

abstracted from *Cancer in North America (2000–2004) Volume One,*[6] based on data collected by cancer registries participating in the SEER program and NPCR. Population data were obtained from the US Census Bureau.[7] Causes of death were coded and classified according to the *International Classification of Diseases* (ICD-8, ICD-9, and ICD-10).[8–10] Cancer cases were classified according to the *International Classification of Diseases for Oncology*.[11]

Estimated New Cancer Cases

The precise number of cancer cases diagnosed each year in the nation and in every state is unknown because complete cancer registration has not yet been achieved in some states. Furthermore, the most recent year for which incidence and mortality data are available lags 3 to 4 years behind the current year due to the time required for data collection and compilation. Estimated new cancer cases in the current year (2008) were projected using a spatio-temporal model[12] based on incidence data from 1995 through 2004 from 41 states and the District of Columbia that met NAACCR's high-quality data standard for incidence, covering about 85% of the US population. The method also considers geographic variations in socio-demographic and lifestyle factors, medical settings, and cancer screening behaviors as predictors of incidence, and accounts for expected delays in case reporting.

Estimated Cancer Deaths

We used the state-space prediction method[13] to estimate the number of cancer deaths expected to occur in the United States and in each state in the year 2008. Projections are based on underlying cause of death from death certificates as reported to the NCHS.[1] This model projects the number of cancer deaths expected to occur in 2008 based on the number that occurred each year from 1969 to 2005 in the United States and in each state separately.

Other Statistics

We provide mortality statistics for the leading causes of death as well as deaths from cancer in the year 2005. Causes of death for 2005 were coded and classified according to ICD-10.[8] This report also provides updated statistics on trends in cancer incidence and mortality rates, the probability of developing cancer, and 5-year relative survival rates for selected cancer sites based on data from 1975 through 2004.[3] All age-adjusted incidence and death rates are standardized to the 2000 US standard population and expressed per 100,000 population.

The long-term incidence rates and trends (1975 to 2004) are adjusted for delays in reporting where possible. Delayed reporting primarily affects the most recent 1 to 3 years of incidence data (in this case, 2002 to 2004), especially for cancers such as melanoma, leukemia, and prostate that are frequently diagnosed in outpatient settings. The National Cancer Institute has developed a method to account for expected reporting delays in SEER registries for all cancer sites combined and many specific cancer sites.[14] Delay-adjusted rates provide a more accurate assessment of trends in the most recent years for which data are available.

We also provide estimates of the total number of cancer deaths avoided due to the reduction in age-standardized cancer death rates since 1991 in men and 1992 in women. We applied the age-specific cancer death rates in the peak year for the age-standardized cancer death rates (1990 for males and 1991 for females) to the corresponding age-specific populations in the subsequent years through 2004 to obtain the number of expected deaths in each calendar year if the death rates had not decreased. We then summed the difference between the number of expected and observed deaths in each age group and calendar year for men and women separately to obtain the total number of cancer deaths avoided over the 13- or 14-year interval.

SELECTED FINDINGS

Expected Numbers of New Cancer Cases

Table 1 presents estimates of the numbers of new cases of invasive cancer expected among men and women in the United States in 2008. The overall estimate of about 1.44 million new cases does not include carcinoma in situ of any site except urinary bladder, nor does it include basal cell and squamous cell cancers of the skin.

TABLE **1** Estimated New Cancer Cases and Deaths by Sex, United States, 2008*

	Estimated New Cases			Estimated Deaths		
	Both Sexes	**Male**	**Female**	**Both Sexes**	**Male**	**Female**
All Sites	**1,437,180**	**745,180**	**692,000**	**565,650**	**294,120**	**271,530**
Oral cavity & pharynx	**35,310**	**25,310**	**10,000**	**7,590**	**5,210**	**2,380**
Tongue	10,140	7,280	2,860	1,880	1,210	670
Mouth	10,820	6,590	4,230	1,840	1,120	720
Pharynx	12,410	10,060	2,350	2,200	1,620	580
Other oral cavity	1,940	1,380	560	1,670	1,260	410
Digestive system	**271,290**	**148,560**	**122,730**	**135,130**	**74,850**	**60,280**
Esophagus	16,470	12,970	3,500	14,280	11,250	3,030
Stomach	21,500	13,190	8,310	10,880	6,450	4,430
Small intestine	6,110	3,200	2,910	1,110	580	530
Colon†	108,070	53,760	54,310	49,960	24,260	25,700
Rectum	40,740	23,490	17,250			
Anus, anal canal, & anorectum	5,070	2,020	3,050	680	250	430
Liver & intrahepatic bile duct	21,370	15,190	6,180	18,410	12,570	5,840
Gallbladder & other biliary	9,520	4,500	5,020	3,340	1,250	2,090
Pancreas	37,680	18,770	18,910	34,290	17,500	16,790
Other digestive organs	4,760	1,470	3,290	2,180	740	1,440
Respiratory system	**232,270**	**127,880**	**104,390**	**166,280**	**94,210**	**72,070**
Larynx	12,250	9,680	2,570	3,670	2,910	760
Lung & bronchus	215,020	114,690	100,330	161,840	90,810	71,030
Other respiratory organs	5,000	3,510	1,490	770	490	280
Bones & joints	**2,380**	**1,270**	**1,110**	**1,470**	**820**	**650**
Soft tissue (including heart)	**10,390**	**5,720**	**4,670**	**3,680**	**1,880**	**1,800**
Skin (excluding basal & squamous)	**67,720**	**38,150**	**29,570**	**11,200**	**7,360**	**3,840**
Melanoma	62,480	34,950	27,530	8,420	5,400	3,020
Other nonepithelial skin	5,240	3,200	2,040	2,780	1,960	820
Breast	**184,450**	**1,990**	**182,460**	**40,930**	**450**	**40,480**
Genital system	**274,150**	**195,660**	**78,490**	**57,820**	**29,330**	**28,490**
Uterine cervix	11,070		11,070	3,870		3,870
Uterine corpus	40,100		40,100	7,470		7,470
Ovary	21,650		21,650	15,520		15,520
Vulva	3,460		3,460	870		870
Vagina & other genital, female	2,210		2,210	760		760
Prostate	186,320	186,320		28,660	28,660	
Testis	8,090	8,090		380	380	
Penis & other genital, male	1,250	1,250		290	290	
Urinary system	**125,490**	**85,870**	**39,620**	**27,810**	**18,430**	**9,380**
Urinary bladder	68,810	51,230	17,580	14,100	9,950	4,150
Kidney & renal pelvis	54,390	33,130	21,260	13,010	8,100	4,910
Ureter & other urinary organs	2,290	1,510	780	700	380	320
Eye & orbit	**2,390**	**1,340**	**1,050**	**240**	**130**	**110**
Brain & other nervous system	**21,810**	**11,780**	**10,030**	**13,070**	**7,420**	**5,650**
Endocrine system	**39,510**	**10,030**	**29,480**	**2,430**	**1,110**	**1,320**
Thyroid	37,340	8,930	28,410	1,590	680	910
Other endocrine	2,170	1,100	1,070	840	430	410
Lymphoma	**74,340**	**39,850**	**34,490**	**20,510**	**10,490**	**10,020**
Hodgkin lymphoma	8,220	4,400	3,820	1,350	700	650
Non-Hodgkin lymphoma	66,120	35,450	30,670	19,160	9,790	9,370
Myeloma	**19,920**	**11,190**	**8,730**	**10,690**	**5,640**	**5,050**
Leukemia	**44,270**	**25,180**	**19,090**	**21,710**	**12,460**	**9,250**
Acute lymphocytic leukemia	5,430	3,220	2,210	1,460	800	660
Chronic lymphocytic leukemia	15,110	8,750	6,360	4,390	2,600	1,790
Acute myeloid leukemia	13,290	7,200	6,090	8,820	5,100	3,720
Chronic myeloid leukemia	4,830	2,800	2,030	450	200	250
Other leukemia‡	5,610	3,210	2,400	6,590	3,760	2,830
Other & unspecified primary sites‡	**31,490**	**15,400**	**16,090**	**45,090**	**24,330**	**20,760**

*Rounded to the nearest 10; estimated new cases exclude basal and squamous cell skin cancers and in situ carcinomas except urinary bladder. About 67,770 female carcinoma in situ of the breast and 54,020 melanoma in situ will be newly diagnosed in 2008.

†Estimated deaths for colon and rectum cancers are combined.

‡More deaths than cases may suggest lack of specificity in recording underlying causes of death on death certificates.

Source: Estimated new cases are based on 1995–2004 incidence data from 41 states and the District of Columbia as reported by the North American Association of Central Cancer Registries (NAACCR), representing about 85% of the US population. Estimated deaths are based on US Mortality Data, 1969 to 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

More than 1 million cases of basal cell and squamous cell skin cancer, about 67,770 cases of breast carcinoma in situ, and 54,020 cases of in situ melanoma are expected to be newly diagnosed in 2008. The estimated numbers of new cancer cases for each state and selected cancer sites are shown in Table 2.

Figure 1 indicates the most common cancers expected to occur in men and women in 2008. Among men, cancers of the prostate, lung and bronchus, and colon and rectum account for about 50% of all newly diagnosed cancers. Prostate cancer alone accounts for about 25% (186,320) of incident cases in men. Based on cases diagnosed between 1996 and 2003, an estimated 91% of these new cases of prostate cancer are expected to be diagnosed at local or regional stages, for which 5-year relative survival approaches 100%.

The 3 most commonly diagnosed types of cancer among women in 2008 will be cancers of the breast, lung and bronchus, and colon and rectum, accounting for about 50% of estimated cancer cases in women. Breast cancer alone is expected to account for 26% (182,460) of all new cancer cases among women.

Expected Number of New Cancer Deaths

Table 1 also shows the expected number of deaths from cancer projected for 2008 for men, women, and both sexes combined. It is estimated that about 565,650 Americans will die from cancer, corresponding to over 1,500 deaths per day. Cancers of the lung and bronchus, prostate, and colon and rectum in men, and cancers of the lung and bronchus, breast, and colon and rectum in women continue to be the most common fatal cancers. These 4 cancers account for half of the total cancer deaths among men and women (Figure 1). Lung cancer surpassed breast cancer as the leading cause of cancer death in women in 1987. Lung cancer is expected to account for 26% of all female cancer deaths in 2008. Table 3 provides the estimated number of cancer deaths in 2008 by state for selected cancer sites.

Regional Variations in Cancer Rates

Table 4 depicts cancer incidence for select cancer sites by state. For some sites, rates vary widely across states. For example, among the cancers listed in Table 4, the largest variations in the incidence rates, in proportionate terms, occurred in lung cancer in which rates (cases per 100,000 population) ranged from 40.3 in men and 20.9 in women in Utah to 137.9 in men and 74.9 in women in Kentucky. In contrast, the variation in female breast cancer incidence rates was small, ranging from 105.4 cases per 100,000 population in Mississippi to 142.2 cases in Washington. Factors that contribute to state variations in incidence rates include differences in the prevalence of risk factors, access to and utilization of early detection services, and completeness of reporting. For example, the state variation in lung cancer incidence rates primarily reflects differences in smoking prevalence. Utah ranks lowest in adult smoking prevalence and Kentucky highest.

Trends in Cancer Incidence and Mortality

Figures 2 to 5 depict long-term trends in cancer incidence and death rates for all cancers combined and for selected cancer sites by sex. Table 5 shows incidence and mortality patterns for all cancer sites and for the 4 most common cancer sites based on joinpoint analysis. Trends in incidence were adjusted for delayed reporting. Delay-adjusted cancer incidence rates were stable in males from 1995 to 2004 and in females from 1999 to 2004. Death rates for all cancer sites combined decreased by 2.6% per year in males and by 1.8% in females from 2002 to 2004 compared with declines of 1.5% per year in males from 1992 to 2002 and 0.8% per year in females from 1994 to 2002.

Mortality rates have continued to decrease across all 4 major cancer sites in both men and women, except for female lung cancer, in which rates continued to increase by 0.2% per year from 1995 to 2004 (Table 5). The incidence trends are mixed, however. Lung cancer incidence rates are declining in men and plateauing in women after increasing for many decades. The lag in the temporal trend of lung cancer incidence rates in women compared with men reflects historical

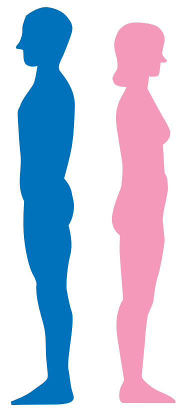
TABLE **2** Age-standardized Incidence Rates for All Cancers Combined, 2000 to 2004, and Estimated New Cases* for Selected Cancers by State, United States, 2008

State	Incidence Rate†	All Cases	Female Breast	Uterine Cervix	Colon & Rectum	Uterine Corpus	Leukemia	Lung & Bronchus	Melanoma of the Skin	Non-Hodgkin Lymphoma	Prostate	Urinary Bladder
AL	437.9	22,340	2,750	170	2,390	490	630	3,900	820	970	2,850	890
AK	477.8	2,650	350	†	250	60	70	340	80	120	450	120
AZ	§	25,540	3,220	200	2,620	610	760	3,850	1,380	1,180	3,610	1,380
AR	§	14,840	1,790	130	1,690	330	520	2,640	540	650	1,980	610
CA	444.0	156,530	20,080	1,280	14,500	4,020	4,530	18,060	7,620	7,560	24,380	6,940
CO	444.5	18,900	2,520	140	1,840	510	720	2,210	1,180	920	3,210	890
CT	503.0	19,190	2,640	120	2,090	660	570	2,680	1,060	910	2,280	1,080
DE	497.1	4,590	580	†	480	130	110	760	180	190	530	220
DC	497.2	2,560	300	†	270	60	50	350	50	100	330	80
FL	472.3	101,920	11,850	770	10,920	2,450	3,190	17,360	4,430	4,750	11,380	5,390
GA	463.7	36,980	4,910	340	3,760	840	1,030	5,980	1,600	1,550	4,700	1,380
HI	416.1	6,310	840	50	700	180	170	710	300	250	930	210
ID	453.7	6,430	760	†	620	160	240	800	360	340	1,160	320
IL	484.3	59,130	7,190	500	6,570	1,790	1,890	9,340	1,930	2,870	7,790	2,840
IN	468.6	29,550	3,660	230	3,350	890	910	5,140	1,130	1,340	3,550	1,380
IA	473.5	16,150	1,990	100	1,810	510	630	2,590	790	730	1,910	830
KS	§	12,520	1,730	80	1,270	390	410	1,910	550	600	1,350	570
KY	508.6	23,270	2,600	190	2,560	580	700	4,580	1,080	970	3,140	990
LA	489.0	23,360	2,840	210	2,490	430	690	3,730	690	1,020	3,430	920
ME	518.3	8,140	990	50	860	270	260	1,330	410	340	1,110	490
MD	§	27,380	3,670	210	2,920	810	640	4,100	1,110	1,080	3,420	1,120
MA	505.9	34,470	4,480	200	3,560	1,120	1,000	4,930	1,810	1,580	3,800	1,950
MI	503.2	51,150	6,120	330	5,150	1,620	1,630	8,020	2,140	2,360	7,180	2,510
MN	§	23,160	3,090	140	2,430	750	910	3,330	830	1,110	3,400	1,110
MS	§	13,400	1,630	130	1,470	230	380	2,290	310	540	2,010	500
MO	460.2	29,390	3,810	210	3,090	860	870	5,560	1,110	1,330	3,050	1,380
MT	472.2	5,090	620	†	530	130	160	710	200	230	900	270
NE	468.8	8,710	1,160	60	910	270	290	1,240	380	390	1,260	420
NV	483.4	11,370	1,270	90	1,160	240	370	1,850	430	480	1,710	570
NH	496.7	7,030	950	†	760	240	200	1,040	400	320	850	390
NJ	513.2	45,900	6,310	370	4,600	1,590	1,440	6,210	2,300	2,210	5,090	2,620
NM	413.5	8,260	1,060	70	830	220	320	940	410	350	1,470	350
NY	483.2	97,130	13,310	830	10,060	3,340	3,140	13,500	3,440	4,460	10,500	5,460
NC	§	40,420	5,000	340	4,380	1,050	1,110	6,510	1,830	1,610	5,050	1,740
ND	436.3	3,090	410	†	350	90	100	390	110	140	480	170
OH	461.2	56,840	6,990	380	6,270	1,830	1,660	9,510	2,110	2,790	6,650	2,810
OK	461.1	17,860	2,270	150	1,860	390	570	3,150	700	840	2,530	750
OR	477.1	19,230	2,430	90	1,740	500	490	2,580	1,120	930	2,730	1,000
PA	497.0	70,110	9,410	440	7,560	2,460	2,220	10,320	3,280	3,300	6,510	4,290
RI	514.0	6,120	770	†	650	200	170	880	310	250	650	370
SC	471.2	20,740	2,510	180	2,170	500	590	3,550	940	780	2,520	850
SD	§	4,080	520	†	430	120	130	500	160	170	580	210
TN	§	29,390	3,720	250	3,290	680	880	5,070	1,150	1,320	3,980	1,250
TX	446.8	96,320	12,210	970	9,570	2,100	3,330	13,840	3,940	4,650	12,960	3,610
UT	407.0	7,760	1,010	60	750	240	320	580	500	420	1,510	340
VT	§	3,530	470	†	360	110	100	460	180	160	490	180
VA	§	35,590	4,680	260	3,690	1,000	850	5,340	1,620	1,410	4,430	1,460
WA	494.9	32,380	4,140	170	2,850	850	970	4,110	1,900	1,590	4,990	1,580
WV	483.6	10,250	1,150	80	1,200	320	290	2,000	440	410	1,180	530
WI	§	27,590	3,400	190	2,930	830	980	3,920	1,010	1,390	3,970	1,360
WY	447.5	2,570	310	†	260	70	80	320	120	110	400	120
US	**471.9**	**1,437,180**	**182,460**	**11,070**	**148,810**	**40,100**	**44,270**	**215,020**	**62,480**	**66,120**	**186,320**	**68,810**

*Rounded to the nearest 10; excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.
†Rates are per 100,000 population and age-adjusted to the 2000 US standard population; source: CINA+ Online, NAACCR, based on data collected by cancer registries participating in NCI's SEER Program and CDC's National Program of Cancer Registries.
‡Estimate is fewer than 50 cases.
§Incidence rate for men and women combined is not available.
Note: These model-based estimates are calculated using incidence data from 41 states and the District of Columbia as reported by NAACCR; they are offered as a rough guide and should be interpreted with caution. State estimates may not add to US total due to rounding and exclusion of states with fewer than 50 cases.

Estimated New Cases*

Males			Females		
Prostate	186,320	25%	Breast	182,460	26%
Lung & bronchus	114,690	15%	Lung & bronchus	100,330	14%
Colon & rectum	77,250	10%	Colon & rectum	71,560	10%
Urinary bladder	51,230	7%	Uterine corpus	40,100	6%
Non-Hodgkin lymphoma	35,450	5%	Non-Hodgkin lymphoma	30,670	4%
Melanoma of the skin	34,950	5%	Thyroid	28,410	4%
Kidney & renal pelvis	33,130	4%	Melanoma of the skin	27,530	4%
Oral cavity & pharynx	25,310	3%	Ovary	21,650	3%
Leukemia	25,180	3%	Kidney & renal pelvis	21,260	3%
Pancreas	18,770	3%	Leukemia	19,090	3%
All Sites	**745,180**	**100%**	**All Sites**	**692,000**	**100%**



Estimated Deaths

Males			Females		
Lung & bronchus	90,810	31%	Lung & bronchus	71,030	26%
Prostate	28,660	10%	Breast	40,480	15%
Colon & rectum	24,260	8%	Colon & rectum	25,700	9%
Pancreas	17,500	6%	Pancreas	16,790	6%
Liver & intrahepatic bile duct	12,570	4%	Ovary	15,520	6%
Leukemia	12,460	4%	Non-Hodgkin lymphoma	9,370	3%
Esophagus	11,250	4%	Leukemia	9,250	3%
Urinary bladder	9,950	3%	Uterine corpus	7,470	3%
Non-Hodgkin lymphoma	9,790	3%	Liver & intrahepatic bile duct	5,840	2%
Kidney & renal pelvis	8,100	3%	Brain & other nervous system	5,650	2%
All Sites	**294,120**	**100%**	**All Sites**	**271,530**	**100%**

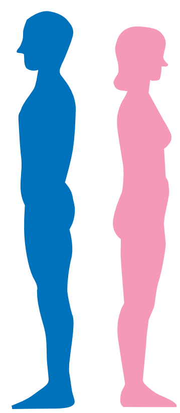


FIGURE **1** Ten Leading Cancer Types for the Estimated New Cancer Cases and Deaths, by Sex, United States, 2008.
*Excludes basal and squamous cell skin cancers and in situ carcinoma except urinary bladder. Estimates are rounded to the nearest 10.

differences in cigarette smoking between men and women; cigarette smoking in women peaked about 20 years later than in men. Colorectal cancer incidence rates have decreased from 1998 through 2004 in both males and females. Female breast cancer incidence rates decreased by 3.5% per year from 2001 to 2004 after increasing since 1980, likely reflecting both delays in diagnosis due to a decrease in mammography utilization and declines in hormone replacement therapy use among postmenopausal women.[15–16] Prostate cancer incidence rates have stabilized from 1995 through 2004 following a short-term rapid increase and subsequent decrease between 1988 and 1995; these trends are thought to reflect changes in utilization of prostate-specific antigen testing.[17–18]

Table 6 shows the contribution of individual cancer sites to the total decrease in overall cancer death rates. Death rates from all cancers combined peaked in 1990 for men and in 1991 for women. Between 1990/1991 and 2004, death rates from cancer decreased by 18.4% among men and by 10.5% among women. Among men, reductions in death rates from lung, prostate, TABLE **3** Age-standardized Death Rates for All Cancers Combined, 2000 to 2004, and Estimated Deaths* from All Cancers Combined and Selected Sites by State, United States, 2008

State	Death Rate†	All Sites	Brain & Other Nervous System	Female Breast	Colon & Rectum	Leukemia	Liver	Lung & Bronchus	Non-Hodgkin Lymphoma	Ovary	Pancreas	Prostate
AL	209.3	9,920	200	730	870	360	310	3,340	320	280	530	490
AK	192.9	810	‡	50	70	‡	‡	230	‡	‡	50	‡
AZ	172.2	10,290	270	700	950	400	370	2,800	340	310	650	640
AR	209.3	6,350	140	410	580	240	200	2,210	190	140	370	360
CA	175.2	55,550	1,500	4,150	5,070	2,170	2,510	13,100	1,910	1,690	3,720	3,400
CO	168.7	6,700	200	530	660	290	220	1,670	200	240	420	350
CT	184.6	6,970	150	480	560	270	210	1,850	230	180	520	400
DE	204.7	1,870	‡	110	150	70	50	590	60	50	110	100
DC	223.5	990	‡	70	90	‡	‡	250	‡	‡	60	70
FL	184.0	41,660	820	2,760	3,420	1,640	1,310	12,490	1,410	1,040	2,400	2,520
GA	199.9	15,040	300	1,110	1,330	540	400	4,570	480	430	850	730
HI	151.0	2,260	‡	140	210	80	120	570	80	50	170	130
ID	174.6	2,470	80	160	200	120	70	630	100	‡	180	170
IL	201.4	23,660	470	1,750	2,250	980	720	6,600	800	650	1,530	1,100
IN	208.2	12,780	250	820	1,130	510	350	3,990	450	360	750	550
IA	187.2	6,480	160	400	570	310	150	1,810	290	190	380	340
KS	187.2	5,360	150	370	520	220	140	1,610	200	150	320	220
KY	225.1	9,500	150	590	840	320	250	3,480	300	210	470	360
LA	223.6	9,350	210	750	920	310	360	2,980	300	220	540	420
ME	207.6	3,270	80	190	260	110	80	980	100	80	200	180
MD	199.5	10,360	220	830	940	390	300	2,920	350	280	660	550
MA	197.4	13,070	270	860	1,100	480	420	3,600	450	360	880	530
MI	196.6	21,210	490	1,310	1,700	790	560	5,890	740	550	1,190	850
MN	183.1	9,100	240	630	760	390	270	2,380	320	260	560	450
MS	215.9	6,010	170	440	590	220	180	2,030	180	150	330	290
MO	203.6	12,630	280	890	1,100	470	360	4,140	460	310	710	460
MT	188.5	1,970	50	130	160	80	‡	580	80	60	110	120
NE	181.4	3,330	90	230	340	150	70	910	130	90	170	210
NV	203.3	4,690	110	340	490	160	160	1,340	110	130	270	240
NH	195.8	2,640	70	190	210	100	70	760	90	60	160	140
NJ	196.7	16,800	330	1,400	1,590	640	540	4,300	550	480	1,060	800
NM	168.6	3,310	80	240	310	120	150	730	110	90	210	200
NY	183.0	34,870	800	2,650	3,140	1,370	1,210	8,990	1,110	1,040	2,340	1,590
NC	199.6	17,450	350	1,300	1,400	600	460	5,470	500	460	1,020	750
ND	177.8	1,220	‡	80	120	‡	‡	330	‡	‡	80	100
OH	206.3	24,410	550	1,800	2,200	900	650	7,350	660	630	1,380	1,350
OK	203.0	7,420	180	510	710	290	200	2,400	200	180	370	290
OR	194.4	7,450	220	510	630	270	210	2,160	380	230	450	380
PA	199.8	29,370	560	2,180	2,560	1,060	830	8,230	1,160	810	1,820	1,430
RI	197.2	2,310	60	140	190	90	60	600	50	60	130	120
SC	204.6	8,860	190	620	730	320	260	2,860	270	210	520	420
SD	185.0	1,620	50	100	160	70	50	450	70	50	100	100
TN	213.6	13,260	350	920	1,130	470	360	4,490	430	360	720	560
TX	189.7	34,960	850	2,520	3,020	1,420	1,680	9,890	1,320	930	2,060	1,730
UT	144.0	2,730	90	240	240	130	80	480	130	90	180	140
VT	187.3	1,140	‡	90	120	50	‡	350	‡	‡	70	70
VA	198.2	13,990	310	1,140	1,260	500	390	4,600	420	390	840	630
WA	190.2	11,370	380	780	940	460	410	3,180	400	360	720	700
WV	216.3	4,580	100	310	450	150	120	1,450	170	130	210	150
WI	187.5	11,220	270	760	910	500	340	2,940	390	310	680	700
WY	183.0	990	‡	60	100	‡	‡	260	‡	‡	60	50
US	**192.7**	**565,650**	**13,070**	**40,480**	**49,960**	**21,710**	**18,410**	**161,840**	**19,160**	**15,520**	**34,290**	**28,660**

*Rounded to the nearest 10.
†Rates are per 100,000 population and age-adjusted to the 2000 US standard population.
‡Estimate is fewer than 50 deaths.
Note: State estimates may not add to US total due to rounding and exclusion of states with fewer than 50 deaths.
Source: US Mortality Data, 1969 to 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

TABLE **4** Cancer Incidence Rates* by Site and State, United States, 2000 to 2004

	All Sites		Breast	Colon & Rectum		Lung & Bronchus		Non-Hodgkin Lymphoma		Prostate	Urinary Bladder	
State	**Male**	**Female**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Male**	**Female**
Alabama †	540.4	370.1	113.9	61.3	42.1	109.7	51.6	19.9	13.7	146.3	30.1	7.4
Alaska †	556.1	416.8	132.0	63.9	50.0	87.0	59.2	24.0	14.6	166.2	41.4	7.8
Arizona	464.1	364.6	114.4	52.3	37.4	71.2	49.3	18.9	13.4	118.6	35.7	8.8
Arkansas	547.1	376.9	117.1	60.5	43.1	113.7	57.8	20.9	15.2	154.6	34.1	8.5
California †	517.3	393.5	126.5	55.0	40.4	69.0	47.9	22.4	15.4	156.9	34.6	8.4
Colorado †	510.2	400.1	129.3	53.1	41.4	65.1	45.9	21.6	16.6	160.1	35.3	9.1
Connecticut †	588.7	445.2	137.3	66.8	49.0	83.0	57.3	24.6	17.1	174.1	44.4	12.0
Delaware †	589.9	430.2	126.1	65.5	46.3	96.6	63.2	23.0	16.4	174.7	40.9	10.4
Dist. of Columbia †	611.5	421.2	134.8	64.0	50.4	90.4	48.8	21.8	12.5	215.0	25.9	8.8
Florida †	553.0	411.2	119.7	60.9	45.3	93.0	61.0	22.1	15.5	147.4	39.7	10.4
Georgia †	568.4	394.5	123.9	61.6	43.9	107.0	53.3	20.1	14.2	165.7	33.2	8.1
Hawaii †	482.0	372.1	124.2	64.7	41.6	67.8	37.6	17.9	13.2	131.8	24.5	5.5
Idaho †	533.6	393.4	123.1	51.9	39.7	71.4	45.8	21.4	17.7	172.3	38.1	8.4
Illinois †	578.2	423.1	126.2	70.0	49.4	94.3	57.0	23.6	16.3	163.4	40.7	10.4
Indiana †	551.7	414.5	121.0	66.6	47.6	106.1	61.8	22.6	16.3	140.0	36.6	9.4
Iowa †	551.6	422.3	125.9	69.0	51.5	89.5	50.8	22.8	17.1	151.1	39.3	9.9
Kansas ‡	-	-	-	-	-	-	-	-	-	-	-	-
Kentucky †	611.2	441.6	122.1	71.3	52.8	137.9	74.9	22.1	16.9	149.2	38.0	9.8
Louisiana †	615.7	402.4	122.0	71.3	48.9	112.3	57.4	22.8	16.0	179.7	35.3	8.5
Maine †	612.4	451.9	130.8	67.2	50.4	100.5	64.5	23.7	17.5	172.0	48.9	13.2
Maryland ‡	-	-	-	-	-	-	-	-	-	-	-	-
Massachusetts †	586.9	452.1	136.7	67.6	49.4	82.9	62.1	23.1	16.9	173.1	45.8	12.8
Michigan †	606.0	431.7	128.8	61.8	45.8	95.0	59.9	24.5	18.1	194.5	41.8	10.5
Minnesota ‡	-	-	-	-	-	-	-	-	-	-	-	-
Mississippi (2002-2004)	546.7	359.8	105.4	63.1	45.2	109.3	49.9	20.1	12.9	158.7	28.5	7.0
Missouri †	535.4	408.9	124.2	65.9	47.5	104.3	60.0	22.0	16.0	132.8	36.1	8.9
Montana †	557.1	407.3	124.5	56.5	43.3	79.8	56.7	22.8	15.0	184.3	41.7	9.9
Nebraska †	550.3	413.9	130.4	69.1	48.6	84.0	47.8	23.8	17.2	160.4	37.9	9.8
Nevada †	555.6	425.8	121.5	59.8	44.6	88.8	71.4	22.0	15.5	158.4	44.7	11.4
New Hampshire †	575.8	440.8	133.9	62.3	47.4	82.1	59.8	25.0	16.5	166.1	46.9	13.2
New Jersey †	613.9	446.4	131.7	70.1	50.8	82.4	55.3	25.6	17.9	192.8	45.8	12.4
New Mexico †	485.4	359.4	112.2	51.5	35.7	59.5	37.7	18.1	13.9	150.7	28.7	7.1
New York †	570.3	427.0	126.0	66.0	49.0	82.0	53.7	23.9	16.7	170.1	41.9	11.2
North Carolina ‡	-	-	-	-	-	-	-	-	-	-	-	-
North Dakota †	518.6	378.8	123.4	66.3	43.3	71.3	43.9	22.1	15.1	175.2	37.1	9.2
Ohio †	542.0	408.9	123.5	64.1	46.8	97.8	58.3	22.7	16.0	149.3	38.7	9.7
Oklahoma †	547.3	403.1	127.0	63.1	44.6	109.2	63.1	22.0	15.7	148.1	33.2	8.2
Oregon †	538.9	433.8	138.8	56.6	42.6	81.4	61.0	24.2	17.3	157.7	40.3	10.2
Pennsylvania †	588.5	436.3	127.1	70.1	50.0	91.8	54.7	24.7	17.1	166.7	44.3	11.5
Rhode Island †	620.2	443.0	129.9	71.1	46.9	97.4	59.5	23.4	16.2	170.6	51.6	14.5
South Carolina †	586.6	391.1	121.0	65.0	45.9	105.1	51.8	20.5	14.8	175.1	33.2	7.5
South Dakota (2001-2004)	577.0	400.9	126.6	66.4	47.5	82.0	43.3	22.5	17.1	191.4	40.7	9.0
Tennessee §	459.2	361.3	113.9	55.9	41.0	100.0	53.6	18.2	13.2	110.8	29.0	7.4
Texas †	535.9	385.1	117.2	59.7	41.0	91.0	51.0	21.7	15.9	147.9	30.4	7.4
Utah †	487.6	345.2	115.7	47.5	35.2	40.3	20.9	23.2	15.7	186.3	29.4	6.5
Vermont ‡	-	-	-	-	-	-	-	-	-	-	-	-
Virginia	511.6	372.8	121.6	57.5	43.2	84.9	50.2	19.3	13.1	157.3	32.5	8.5
Washington †	567.1	444.1	142.2	55.9	42.5	82.0	60.5	26.6	18.3	172.6	41.7	10.3
West Virginia †	570.6	426.5	115.5	71.7	53.2	116.8	67.7	21.4	16.1	144.2	39.9	11.2
Wisconsin	543.4	413.4	129.0	62.3	45.4	81.3	51.5	22.3	16.5	163.9	36.5	10.2
Wyoming †	519.2	392.2	121.6	49.5	43.4	65.3	44.8	18.9	17.6	179.9	38.9	9.5
United States	557.8	413.1	125.3	62.9	45.8	89.0	55.2	22.8	16.2	160.8	38.4	9.8

*Per 100,000 population, age-adjusted to the 2000 US standard population.

†This state's registry has submitted 5 years of data and passed rigorous criteria for each single year's data, including completeness of reporting, non-duplication of records, percent unknown in critical data fields, percent of cases registered with information from death certificates only, and internal consistency among data items.

‡This state's registry did not submit incidence data to the North American Association of Central Cancer Registries (NAACCR) for 2000 to 2004.

§Case ascertainment for this state's registry is incomplete for the years 2000 to 2004.

Source: CINA+ Online and Cancer in North America: 2000–2004, Volume One: Incidence, NAACCR, 2007. Data are collected by cancer registries participating in the National Cancer Institute's SEER program and the Centers for Disease Control and Prevention's National Program of Cancer Registries.

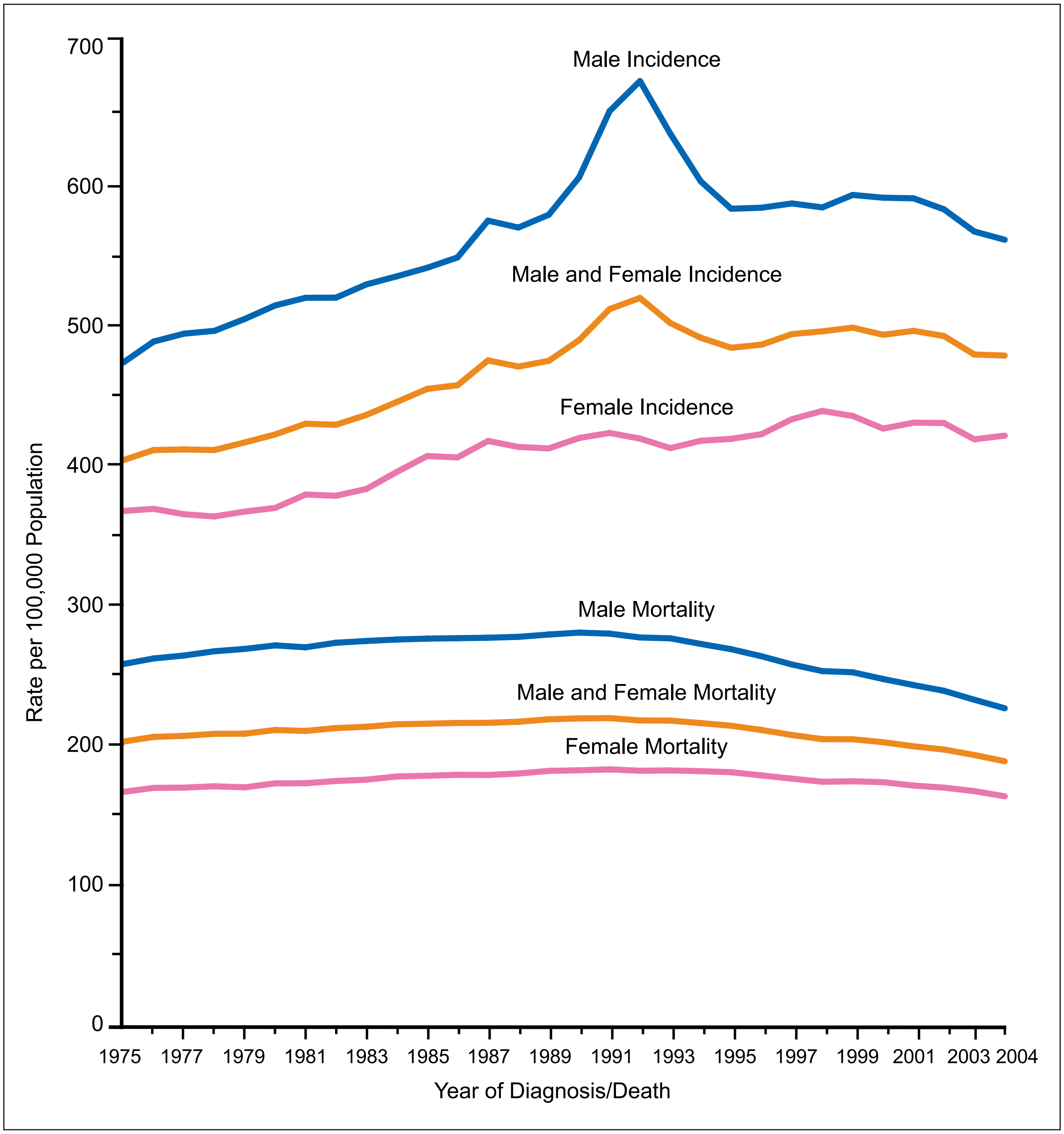


FIGURE **2** Annual Age-adjusted Cancer Incidence and Death Rates* for All Sites by Sex, United States, 1975 to 2004.

*Rates are age-adjusted to the 2000 US standard population. Incidence rates are adjusted for delays in reporting.
Source: Incidence—Surveillance, Epidemiology, and End Results (SEER) program, (www.seer.cancer.gov). Delay-Adjusted Incidence database: “SEER Incidence Delay-Adjusted Rates, 9 Registries, 1975–2004.” National Cancer Institute, DCCPS, Surveillance Research Program, Statistical Research and Applications Branch, released April 2007, based on the November 2006 SEER data submission. Mortality—US Mortality Data, 1960 to 2004, National Center for Health Statistics, Centers for Disease Control and Prevention, 2006.

and colorectal cancers account for nearly 80% of the total decrease in cancer death rates, while reductions in death rates from breast and colorectal cancers account for 60% of the decrease among women. Lung cancer in men and breast cancer in women alone account for nearly 40% of the sex-specific decreases in cancer death rates. The decrease in lung cancer death rates among men is due to reduction in tobacco use over the past 40 years, while the decrease in death rates for colorectal, female breast, and prostate cancer largely reflects improvements in early detection and treatment.[18] Between 1990/1991 and 2004, death rates increased substantially for lung

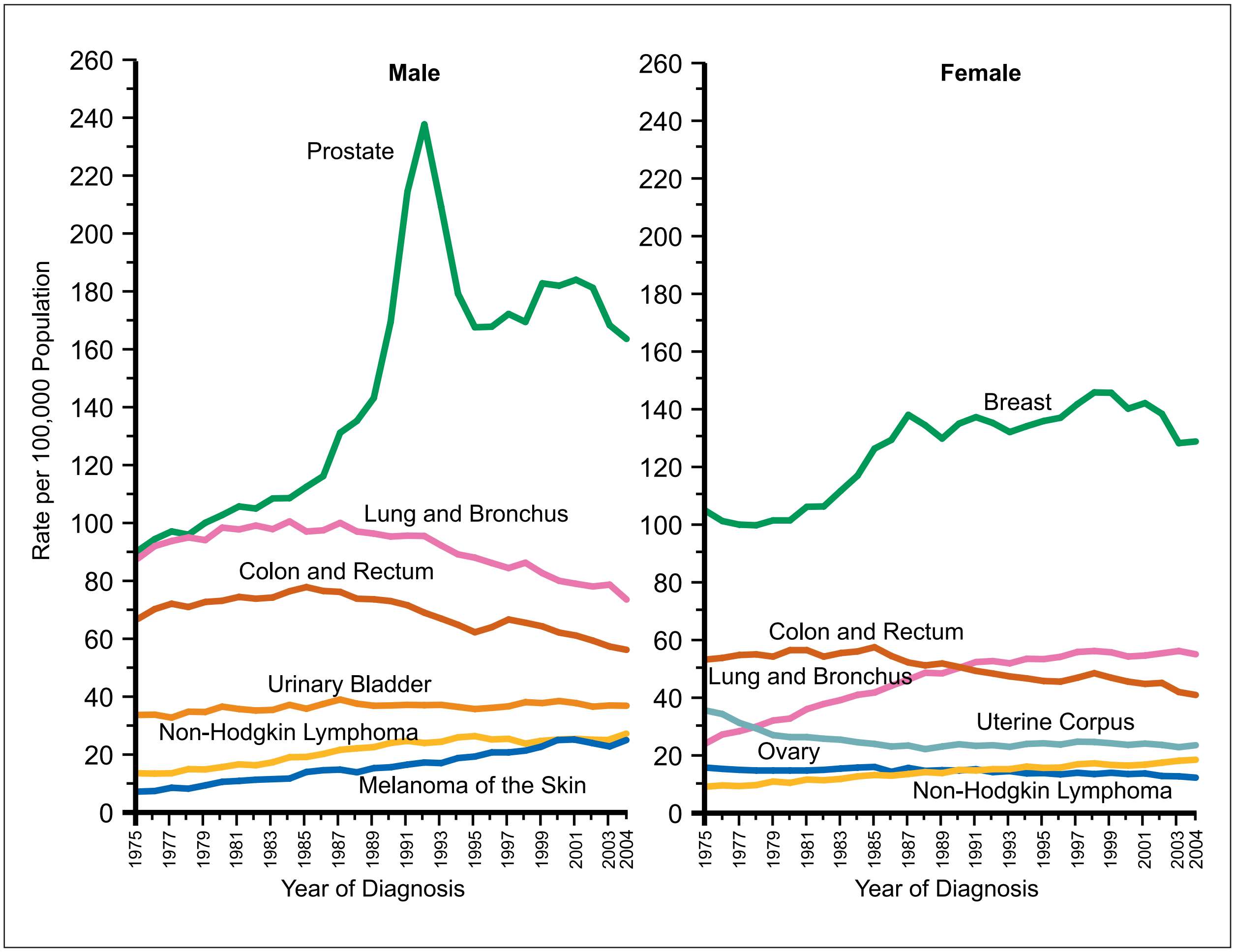


FIGURE **3** Annual Age-adjusted Cancer Incidence Rates* for Selected Cancers by Sex, United States, 1975 to 2004.
*Rates are age-adjusted to the 2000 US standard population and adjusted for delays in reporting.
Source: Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov). Delay-Adjusted Incidence database: "SEER Incidence Delay-Adjusted Rates, 9 Registries, 1975–2004." National Cancer Institute, DCCPS, Surveillance Research Program, Statistical Research and Applications Branch, released April 2007, based on the November 2006 SEER data submission.

cancer in women and for liver and intrahepatic bile duct cancer in men.

Recorded Number of Deaths from Cancer in 2005

A total of 559,312 cancer deaths were recorded in the United States in 2005, the most recent year for which actual data are available, accounting for about 23% of all deaths (Table 7). Despite a decrease in age-standardized death rates, there were 5,424 more cancer deaths reported in 2005 than in 2004 (3,592 in men and 1,832 in women) (Table 8). This contrasts with the decrease in the total number of cancer deaths that occurred from 2002 to 2003 and 2003 to 2004, in part due to a larger decrease in the age-standardized death rates from 2002 to 2004 (about 2% per year) compared with 2004 to 2005 (about 1% per year). With respect to the 4 major cancer sites, colorectal cancer death rates decreased by about 6% from 2003 to 2004 compared with only about 3% from 2004 to 2005. The declines in death rates for cancers of the lung and bronchus and prostate in men and breast in women were similarly attenuated from 2004 to 2005.

Cancer is one of the 5 leading causes of death in all age groups among both males and females (Table 9). Cancer is the leading cause of death among women aged 40 to 79 years and among men aged 60 to 79 years. Cancer is the leading cause of death among men and women under age 85 years (Figure 6). A total of 475,848 people under age 85 years died from cancer in the

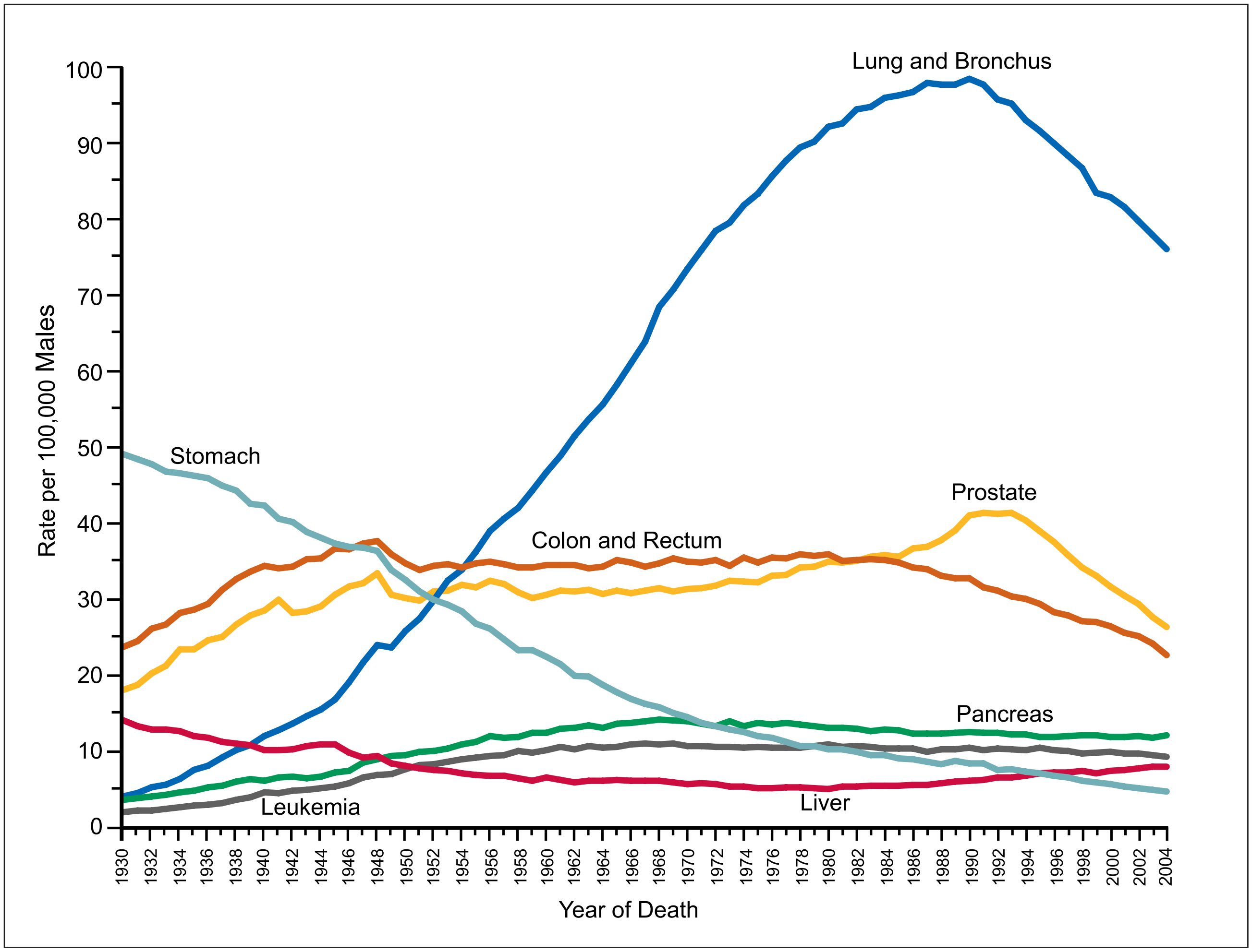


FIGURE **4** Annual Age-adjusted Cancer Death Rates* Among Males for Selected Cancers, United States, 1930 to 2004.
*Rates are age-adjusted to the 2000 US standard population.
Note: Due to changes in ICD coding, numerator information has changed over time. Rates for cancers of the lung and bronchus, colon and rectum, and liver are affected by these changes.
Source: US Mortality Data, 1960 to 2004, US Mortality Volumes, 1930 to 1959, National Center for Health Statistics, Centers for Disease Control and Prevention, 2006.

United States in 2005 compared with 408,550 deaths from heart disease.

Table 10 presents the number of deaths from the 5 most common cancer sites for males and females at various ages. Among males under age 40 years, leukemia is the most common fatal cancer, while cancer of the lung and bronchus predominates in men aged 40 years and older. Among females, leukemia is the leading cause of cancer death before age 20 years, breast cancer ranks first at age 20 to 59 years, and lung cancer ranks first at age 60 years and older.

Figure 7 shows the total number of cancer deaths avoided since death rates began to decrease in 1991 in men and in 1992 in women. As a result, over a half million cancer deaths (408,400 in men and 136,100 in women) were averted during the time interval of 1991/1992 through 2004.

CANCER OCCURRENCE BY RACE/ETHNICITY

Cancer incidence and death rates vary considerably among racial and ethnic groups (Table 11). For all cancer sites combined, African American men have a 19% higher incidence rate and a 37% higher death rate than White men, whereas African American women have a 6% lower incidence rate but a 17% higher death rate than White women. For the specific cancer sites

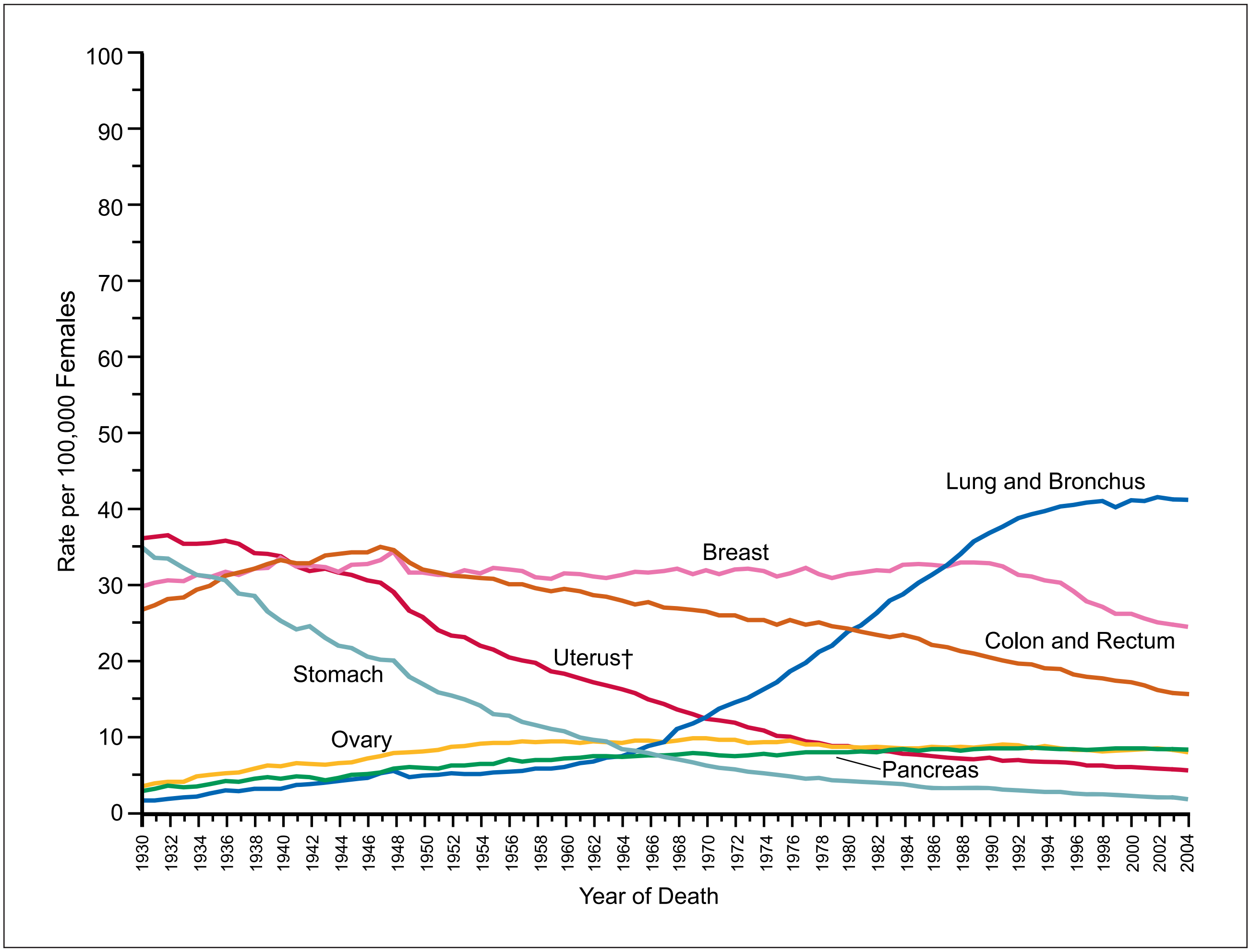


FIGURE **5** Annual Age-adjusted Cancer Death Rates* Among Females for Selected Cancers, United States, 1930 to 2004.
*Rates are age-adjusted to the 2000 US standard population.
†Uterus includes uterine cervix and uterine corpus.
Note: Due to changes in ICD coding, numerator information has changed over time. Rates for cancers of the uterus, ovary, lung and bronchus, and colon and rectum are affected by these changes.
Source: US Mortality Data, 1960 to 2004, US Mortality Volumes 1930 to 1959, National Center for Health Statistics, Centers for Disease Control and Prevention, 2006.

listed in Table 11, incidence and death rates are consistently higher in African Americans than in Whites, except for cancers of the breast (incidence) and lung (incidence and mortality) among women and kidney (mortality) among both men and women. Factors known to contribute to racial disparities in mortality vary by cancer site and include differences in exposure to underlying risk factors (eg, historical smoking prevalence for lung cancer among men), access to high-quality regular screening (breast, cervical, and colorectal cancers), and timely diagnosis and treatment (for many cancers). The higher breast cancer incidence rates among Whites are thought to reflect a combination of factors that affect both diagnosis (such as more frequent mammography in White women) and the underlying factors that affect disease occurrence (such as later age at first birth and greater use of hormone replacement therapy among White compared with African American women).[19]

Among other racial and ethnic groups, cancer incidence and death rates are lower than those in Whites and African Americans for all cancer sites combined and for the 4 most common cancer sites. However, incidence and death rates are generally higher in minority populations than in Whites for cancers of the uterine cervix, TABLE **5** Trends in Cancer Incidence and Death Rates for Selected Cancers by Sex, United States, 1975 to 2004

	Line Segment 1		**Line Segment 2**		**Line Segment 3**		**Line Segment 4**	
	Year	**APC***	**Year**	**APC***	**Year**	**APC***	**Year**	**APC***
All sites								
Incidence								
Male and female	1975-1992	1.4 †	1992-2004	-0.3 †				
Male	1975-1989	1.3 †	1989-1992	5.1 †	1992-1995	-4.3 †	1995-2004	-0.2
Female	1975-1979	-0.3	1979-1986	1.6 †	1986-1999	0.4 †	1999-2004	-0.5
Death								
Male and female	1975-1990	0.5 †	1990-1993	-0.3	1993-2002	-1.1 †	2002-2004	-2.1 †
Male	1975-1980	0.9 †	1980-1992	0.3 †	1992-2002	-1.5 †	2002-2004	-2.6 †
Female	1975-1990	0.6 †	1990-1994	-0.2	1994-2002	-0.8 †	2002-2004	-1.8 †
Lung & bronchus								
Incidence								
Male and female	1975-1982	2.5 †	1982-1991	1.0 †	1991-2004	-0.7 †		
Male	1975-1982	1.5 †	1982-1991	-0.5	1991-2004	-1.8 †		
Female	1975-1982	5.5 †	1982-1990	3.5 †	1990-1998	1.0 †	1998-2004	-0.1
Death								
Male and female	1975-1980	3.0 †	1980-1990	1.8 †	1990-1995	-0.2	1995-2004	-1.0 †
Male	1975-1982	1.7 †	1982-1990	0.5 †	1990-1994	-1.3 †	1994-2004	-2.0 †
Female	1975-1982	6.0 †	1982-1990	4.2 †	1990-1995	1.7 †	1995-2004	0.2 †
Colon & rectum								
Incidence								
Male and female	1975-1985	0.8 †	1985-1995	-1.8 †	1995-1998	1.3	1998-2004	-2.3 †
Male	1975-1986	1.1 †	1986-1995	-2.1 †	1995-1998	1.1	1998-2004	-2.6 †
Female	1975-1985	0.3	1985-1995	-1.8 †	1995-1998	1.8	1998-2004	-2.2 †
Death								
Male and female	1975-1978	0.2	1978-1985	-0.8 †	1985-2002	-1.8 †	2002-2004	-4.7 †
Male	1975-1979	0.6	1979-1987	-0.6 †	1987-2002	-1.9 †	2002-2004	-4.9 †
Female	1975-1984	-1.0 †	1984-2002	-1.8 †	2002-2004	-4.5 †		
Breast (female)								
Incidence	1975-1980	-0.4	1980-1987	3.7 †	1987-2001	0.5 †	2001-2004	-3.5 †
Death	1975-1990	0.4 †	1990-2004	-2.2 †				
Prostate								
Incidence	1975-1988	2.6 †	1988-1992	16.2 †	1992-1995	-10.2 †	1995-2004	0.4
Death	1975-1987	0.9 †	1987-1991	3.0 †	1991-1994	-0.5	1994-2004	-4.1 †

*Annual percent change based on incidence (delay-adjusted) and mortality rates age-adjusted to the 2000 US standard population.
†The APC is significantly different from zero.
Note: Trends were analyzed by Joinpoint Regression Program, Version 3.0, with a maximum of 3 joinpoints (ie, 4 line segments).
Source: Ries LAG, Melbert D, Krapcho M, et al.[3]

stomach, and liver. Stomach and liver cancer incidence and death rates are more than twice as high in Asian American/Pacific Islanders as in Whites, reflecting increased exposure to infectious agents such as *Helicobacter pylori* and hepatitis B and C.[20] Kidney cancer mortality rates are the highest among American Indians/Alaska Natives, although obesity is the only factor known to contribute to this disparity.

Trends in cancer incidence can be adjusted for delayed reporting only in Whites and African Americans because long-term incidence data required for delay adjustment are not available for other racial and ethnic subgroups. From 1995 to 2004, incidence rates in males for all cancer sites combined (unadjusted for delayed reporting) decreased among all racial and ethnic groups except American Indian/Alaska Native men; among women, rates stabilized in all racial and ethnic groups during the same time period. In contrast, death rates from cancer significantly decreased in each racial and ethnic group, with larger decreases in men than in women.[3]

Death Rates by Educational Attainment, Race, and Sex

Table 12 shows death rates for all cancers combined and the 4 major cancers by educational attainment among White and African American men and women aged 25 to 64 years in 2001.[21] The death rate for all cancers combined among less-educated (≤12 years of education) compared with more-educated (>12 years of

TABLE **6** The Contribution of Indvidual Cancer Sites to the Decrease in Cancer Death Rates, 1990 to 2004

	Death Rate (per 100,000)		Change		% Contribution†
	1990*	**2004**	**Absolute**	**%**	
Male					
All malignant cancers	279.82	228.26	-51.56	-18.43	
Decreasing					
Lung & bronchus	90.56	70.29	-20.27	-22.38	37.3
Prostate	38.56	25.45	-13.11	-34.00	24.1
Colon & rectum	30.77	21.57	-9.20	-29.90	16.9
Pancreas	12.59	12.34	-0.25	-1.99	0.5
Leukemia	10.71	9.70	-1.01	-9.43	1.9
Non-Hodgkin lymphoma	9.97	8.85	-1.12	-11.23	2.1
Urinary bladder	7.97	7.58	-0.39	-4.89	0.7
Kidney & renal pelvis	6.16	5.91	-0.25	-4.06	0.5
Stomach	8.86	5.53	-3.33	-37.58	6.1
Brain & other nervous system	5.97	5.22	-0.75	-12.56	1.4
Myeloma	4.83	4.40	-0.43	-8.90	0.8
Oral cavity & pharynx	5.61	3.98	-1.63	-29.06	3.0
Larynx	2.97	2.24	-0.73	-24.58	1.3
Soft tissue including heart	1.52	1.44	-0.08	-5.26	0.1
Hodgkin lymphoma	0.85	0.54	-0.31	-36.47	0.6
Bones & joints	0.55	0.52	-0.03	-5.45	0.1
Gallbladder	0.60	0.48	-0.12	-20.00	0.2
Other	24.54	23.15	-1.39	-5.66	2.6
Total			**-54.40**		**100.0**
Increasing					
Esophagus	7.16	7.72	0.56	7.82	19.7
Liver & intrahepatic bile duct	5.27	7.41	2.14	40.61	75.4
Melanoma of the skin	3.80	3.94	0.14	3.68	4.9
Total			**2.84**		**100.0**
	Death Rate (per 100,000)		**Change**		**% Contribution†**
	1991*	**2004**	**Absolute**	**%**	
Females					
All malignant cancers	175.30	156.96	-18.34	-10.46	
Decreasing					
Breast	32.69	24.38	-8.31	-25.42	37.3
Colon & rectum	20.30	15.15	-5.15	-25.37	23.1
Pancreas	9.28	9.24	-0.04	-0.43	0.2
Ovary	9.51	8.75	-0.76	-7.99	3.4
Non-Hodgkin lymphoma	6.74	5.68	-1.06	-15.73	4.8
Leukemia	6.32	5.46	-0.86	-13.61	3.9
Corpus & uterus	4.18	4.12	-0.06	-1.44	0.3
Brain & other nervous system	4.11	3.52	-0.59	-14.36	2.6
Myeloma	3.26	2.96	-0.30	-9.20	1.3
Stomach	4.01	2.84	-1.17	-29.18	5.2
Kidney & renal pelvis	2.95	2.72	-0.23	-7.80	1.0
Cervix uteri	3.49	2.41	-1.08	-30.95	4.8
Urinary bladder	2.34	2.24	-0.10	-4.27	0.4
Esophagus	1.81	1.72	-0.09	-4.97	0.4
Melanoma of the skin	1.82	1.70	-0.12	-6.59	0.5
Oral cavity & pharynx	2.03	1.47	-0.56	-27.59	2.5
Soft tissue including heart	1.28	1.12	-0.16	-12.50	0.7
Gallbladder	1.09	0.78	-0.31	-28.44	1.4
Others	17.97	16.62	-1.35	-7.51	6.1
Total			**-22.30**		**100.0**
Increasing					
Lung & bronchus	37.61	40.87	3.26	8.67	82.3
Liver & intrahepatic bile duct	2.51	3.21	0.70	27.89	17.7
Total			**3.96**		**100.0**

*Death rates from cancer peaked in 1990 for men and in 1991 for women.

†This calculation is based on each cancer site's contribution to the increasing or decreasing portion of the total cancer death rate, depending on the individual site's trend; it does not represent the contribution to the net decrease in cancer death rates.

TABLE **7** Fifteen Leading Causes of Death, United States, 2005

Rank	Cause of Death	Number of Deaths	Percent (%) of Total Deaths	Death Rate*
	All Causes	**2,448,017**	**100.0**	**798.8**
1	Heart diseases	652,091	26.6	211.1
2	Cancer	559,312	22.8	183.8
3	Cerebrovascular diseases	143,579	5.9	46.6
4	Chronic lower respiratory diseases	130,933	5.3	43.2
5	Accidents (unintentional injuries)	117,809	4.8	39.1
6	Diabetes mellitus	75,119	3.1	24.6
7	Alzheimer disease	71,599	2.9	22.9
8	Influenza & pneumonia	63,001	2.6	20.3
9	Nephritis, nephrotic syndrome, & nephrosis	43,901	1.8	14.3
10	Septicemia	34,136	1.4	11.2
11	Intentional self-harm (suicide)	32,637	1.3	10.9
12	Chronic liver disease & cirrhosis	27,530	1.1	9.0
13	Hypertension & hypertensive renal disease	24,902	1.0	8.0
14	Parkinson disease	19,544	0.8	6.4
15	Assault (homicide)	18,124	0.7	6.1
	All other & ill-defined causes	433,800	17.7	

*Rates are per 100,000 population and age-adjusted to the 2000 US standard population.
Note: Percentages may not total 100 due to rounding. "Symptoms, signs, and abnormal clinical or laboratory findings" and "Other respiratory diseases" were excluded from the cause of death ranking order.
Source: US Mortality Data, 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

TABLE **8** Trends in the Recorded Number of Deaths for Selected Cancers by Sex, United States, 1990 to 2005

	All Sites		Lung and Bronchus		Colon and Rectum		Prostate	Breast
Year	Male	Female	Male	Female	Male	Female	Male	Female
1990	268,283	237,039	91,014	50,136	28,484	28,674	32,378	43,391
1991	272,380	242,277	91,603	52,022	28,026	28,753	33,564	43,583
1992	274,838	245,740	91,322	54,485	28,280	28,714	34,240	43,068
1993	279,375	250,529	92,493	56,234	28,199	29,206	34,865	43,555
1994	280,465	253,845	91,825	57,535	28,471	28,936	34,902	43,644
1995	281,611	256,844	91,800	59,304	28,409	29,237	34,475	43,844
1996	281,898	257,635	91,559	60,351	27,989	28,766	34,123	43,091
1997	281,110	258,467	91,278	61,922	28,075	28,621	32,891	41,943
1998	282,065	259,467	91,399	63,075	28,024	28,950	32,203	41,737
1999	285,832	264,006	89,401	62,662	28,313	28,909	31,729	41,144
2000	286,082	267,009	90,415	65,016	28,484	28,950	31,078	41,872
2001	287,075	266,693	90,367	65,606	28,229	28,579	30,719	41,394
2002	288,768	268,503	90,121	67,509	28,472	28,132	30,446	41,514
2003	287,990	268,912	89,908	68,084	27,991	27,793	29,554	41,620
2004	286,830	267,058	89,575	68,431	26,881	26,699	29,002	40,954
2005	290,422	268,890	90,141	69,079	26,783	26,224	28,905	41,116

Note: Effective with the mortality data for 1999, causes of death are classified by ICD-10, replacing ICD-9 used for 1979 to 1998 data.
Source: US Mortality Data, 1990 to 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

education) people was more than twice as high in men and about 40% higher in women. For specific cancer sites, the ratio of death rates in the less-educated compared with the more-educated group ranged from 1.16 for breast cancer among African American women to 3.36 for lung cancer among White men. For lung cancer in men and women and for colorectal cancer in men, the absolute difference in death rates between the less educated and more educated was larger than the difference between Whites and African Americans by sex at each

TABLE **9** Ten Leading Causes of Death by Age and Sex, United States, 2005

	All Ages		Ages 1 to 19		Ages 20 to 39		Ages 40 to 59		Ages 60 to 79		Ages 80+	
	Male	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**
	All Causes 1,207,675	**All Causes 1,240,342**	**All Causes 16,504**	**All Causes 8,557**	**All Causes 64,917**	**All Causes 28,959**	**All Causes 225,758**	**All Causes 138,615**	**All Causes 469,362**	**All Causes 384,704**	**All Causes 414,917**	**All Causes 667,029**
1	Heart diseases 322,841	Heart diseases 329,250	Accidents (unintentional injuries) 7,165	Accidents (unintentional injuries) 3,530	Accidents (unintentional injuries) 23,194	Accidents (unintentional injuries) 7,359	Heart diseases 54,601	Cancer 51,059	Cancer 152,720	Cancer 126,469	Heart diseases 134,981	Heart diseases 218,373
2	Cancer 290,422	Cancer 268,890	Assault (homicide) 2,233	Cancer 892	Intentional self-harm (suicide) 8,520	Cancer 4,725	Cancer 53,892	Heart diseases 21,795	Heart diseases 127,021	Heart diseases 86,078	Cancer 78,304	Cancer 85,711
3	Accidents (unintentional injuries) 76,375	Cerebrovascular disease 86,993	Intentional self-harm (suicide) 1,507	Assault (homicide) 559	Assault (homicide) 8,137	Heart diseases 2,519	Accidents (unintentional injuries) 23,999	Accidents (unintentional injuries) 10,008	Chronic lower respiratory diseases 31,879	Chronic lower respiratory diseases 30,607	Cerebrovascular diseases 27,502	Cerebrovascular diseases 58,740
4	Chronic lower respiratory diseases 62,435	Chronic lower respiratory diseases 68,498	Cancer 1,216	Congenital anomalies 540	Heart diseases 5,536	Intentional self-harm (suicide) 1,962	Intentional self-harm (suicide) 9,961	Cerebrovascular diseases 5,654	Cerebrovascular diseases 21,387	Cerebrovascular diseases 21,800	Chronic lower respiratory diseases 25,356	Alzheimer disease 43,850
5	Cerebrovascular diseases 56,586	Alzheimer disease 51,040	Congenital anomalies 631	Intentional self-harm (suicide) 378	Cancer 4,240	Assault (homicide) 1,587	Chronic liver disease & cirrhosis 9,638	Diabetes mellitus 4,781	Diabetes mellitus 17,438	Diabetes mellitus 16,114	Influenza & pneumonia 16,286	Chronic lower respiratory diseases 32,949
6	Diabetes mellitus 36,538	Accidents (unintentional injuries) 41,434	Heart diseases 480	Heart diseases 312	HIV disease 2,073	HIV disease 1,069	Diabetes mellitus 7,343	Chronic lower respiratory diseases 4,545	Accidents (unintentional injuries) 11,961	Nephritis, nephrotic syndrome & nephrosis 7,732	Alzheimer disease 15,599	Influenza & pneumonia 25,421
7	Influenza & pneumonia 28,052	Diabetes mellitus 38,581	Influenza & pneumonia 158	Influenza & pneumonia 126	Diabetes mellitus 903	Cerebrovascular disease 643	Cerebrovascular diseases 6,774	Chronic liver disease & cirrhosis 3,843	Influenza & pneumonia 8,452	Accidents (unintentional injuries) 7,482	Diabetes mellitus 10,799	Diabetes mellitus 17,022
8	Intentional self-harm (suicide) 25,907	Influenza & pneumonia 34,949	Chronic lower respiratory diseases 139	Cerebrovascular disease 103	Cerebrovascular disease 716	Diabetes mellitus 611	HIV disease 6,260	Intentional self-harm (suicide) 3,164	Nephritis, nephrotic syndrome & nephrosis 8,242	Influenza & pneumonia 7,194	Nephritis, nephrotic syndrome & nephrosis 10,012	Nephritis, nephrotic syndrome & nephrosis 12,736
9	Nephritis, nephrotic syndrome & nephrosis 21,268	Nephritis, nephrotic syndrome & nephrosis 22,633	Cerebrovascular disease 130	Septicemia 98	Chronic liver disease & cirrhosis 685	Pregnancy, childbirth & puerperium 586	Chronic lower respiratory diseases 4,733	Septicemia 2,147	Septicemia 6,433	Alzheimer disease 7,025	Accidents (unintentional injuries) 9,429	Accidents (unintentional injuries) 12,542
10	Alzheimer disease 20,559	Septicemia 18,814	Septicemia 129	In situ & benign neoplasms 83	Congenital anomalies 527	Congenital anomalies 393	Assault (homicide) 3,095	HIV disease 2,020	Chronic liver disease & cirrhosis 6,402	Septicemia 6,445	Parkinson disease 6,987	Hypertension & hypertensive renal disease 10,633

Note: Deaths within each age group do not sum to all ages combined due to the inclusion of unknown ages. “Symptoms, signs, and abnormal clinical and laboratory findings,” “Events of undetermined intent,” and “Other respiratory diseases” were excluded from the cause of death ranking order.
Source: US Mortality Data, 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

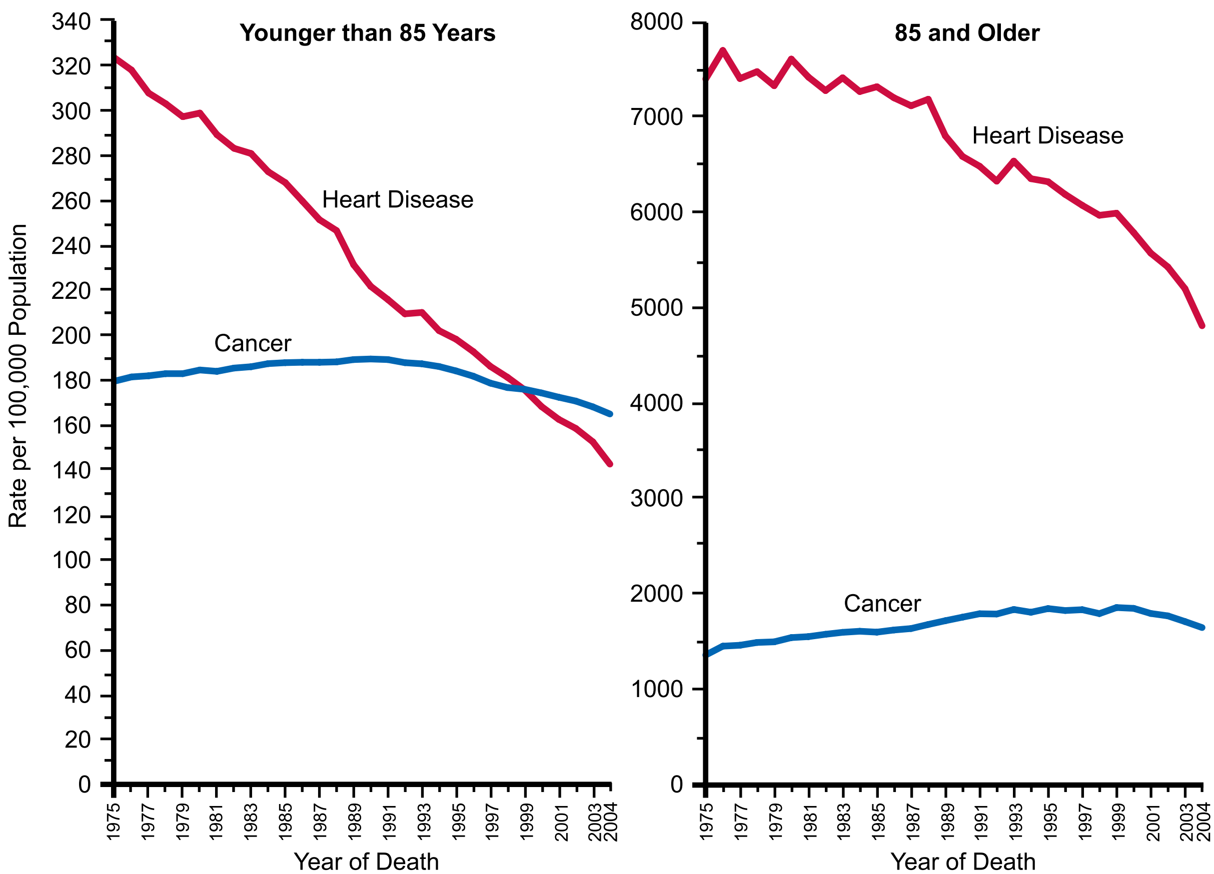


FIGURE 6 Death Rates* for Cancer and Heart Disease for Ages Younger than 85 and 85 and Older, 1975 to 2004.
*Rates are age-adjusted to the 2000 US standard population.
Source: US Mortality Data, 1960 to 2004, National Center for Health Statistics, Centers for Disease Control and Prevention, 2006.

level of educational attainment. Factors that contribute to higher death rates from cancer in less-educated men and women include higher prevalence of risk factors such as smoking and obesity and limited access to medical services.

Lifetime Probability of Developing Cancer

The lifetime probability of being diagnosed with an invasive cancer is higher for men (45%) than women (38%) (Table 13). However, because of the relatively early age of breast cancer onset, women have a slightly higher probability of developing cancer before age 60 years. It is noteworthy that these estimates are based on the average experience of the general population and may over- or underestimate individual risk because of differences in exposure and/or genetic susceptibility.

Cancer Survival by Race

Compared with Whites, African American men and women have poorer survival once cancer is diagnosed. Five-year relative survival is lower in African Americans than in Whites within each stratum of stage of diagnosis for nearly every cancer site (Figure 8). These disparities may result from inequalities in access to and receipt of quality health care and/or from differences in comorbidities.[18] As shown in Figure 9, African Americans are less likely than Whites to be diagnosed with cancer at a localized stage, when the disease may be more easily

TABLE **10** Reported Deaths for the Five Leading Cancer Sites by Age and Sex, United States, 2005

All Ages	<20	20 to 39	40 to 59	60 to 79	≥ 80
Male					
All Sites 290,422	**All Sites 1,261**	**All Sites 4,240**	**All Sites 53,892**	**All Sites 152,720**	**All Sites 78,304**
Lung & bronchus 90,141	Leukemia 382	Leukemia 571	Lung & bronchus 15,939	Lung & bronchus 54,692	Lung & bronchus 19,185
Prostate 28,905	Brain & ONS 298	Brain & ONS 524	Colon & rectum 4,994	Colon & rectum 13,525	Prostate 15,341
Colon & rectum 26,783	Bones & joints 121	Colon & rectum 400	Liver & bile duct 3,594	Prostate 12,378	Colon & rectum 7,850
Pancreas 16,147	Other endocrine system 109	Lung & bronchus 316	Pancreas 3,517	Pancreas 8,772	Urinary bladder 3,985
Leukemia 12,273	Non-Hodgkin lymphoma 78	Non-Hodgkin lymphoma 305	Esophagus 2,746	Leukemia 5,934	Pancreas 3,734
Female					
All Sites 268,890	**All Sites 922**	**All Sites 4,725**	**All Sites 51,059**	**All Sites 126,469**	**All Sites 85,711**
Lung & bronchus 69,079	Leukemia 258	Breast 1,170	Breast 12,198	Lung & bronchus 39,811	Lung & bronchus 17,648
Breast 41,116	Brain & ONS 246	Leukemia 421	Lung & bronchus 11,349	Breast 16,914	Colon & rectum 11,634
Colon & rectum 26,224	Bones & joints 91	Uterine cervix 416	Colon & rectum 3,791	Colon & rectum 10,468	Breast 10,834
Pancreas 16,613	Soft tissue 87	Colon & rectum 325	Ovary 3,448	Pancreas 7,934	Pancreas 6,217
Ovary 14,787	Other endocrine system 69	Brain & ONS 309	Pancreas 2,386	Ovary 7,231	Non-Hodgkin lymphoma 4,139

Abbreviation: ONS, other nervous system.
Note: Deaths within each age group do not sum to all ages combined due to the inclusion of unknown ages. Others and Unspecified Primary are excluded from cause of death ranking order.
Source: US Mortality Data, 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

and successfully treated, and are more likely to be diagnosed with cancer at a regional or distant stage of disease. The extent to which factors other than stage at diagnosis contribute to the overall differential survival is unclear.[22] However, some studies suggest that African Americans who receive cancer treatment and medical care similar to that of Whites experience similar outcomes.[23]

There have been notable improvements since 1975 in relative 5-year survival rates for many cancer sites and for all cancers combined (Table 14). This is true for both Whites and African Americans. Cancers for which survival has not improved substantially over the past 25 years include uterine corpus, cervix, larynx, lung, and pancreas. The improvement in survival reflects a combination of earlier diagnosis and improved treatments.

Relative survival rates cannot be calculated for racial and ethnic populations other than Whites and African Americans because accurate life expectancies (the average number of years of life remaining for persons who have attained a given age) are not available. However, based on cause-specific survival rates of cancer patients diagnosed from 1992 to 2000 in SEER areas of the United States, all minority populations except

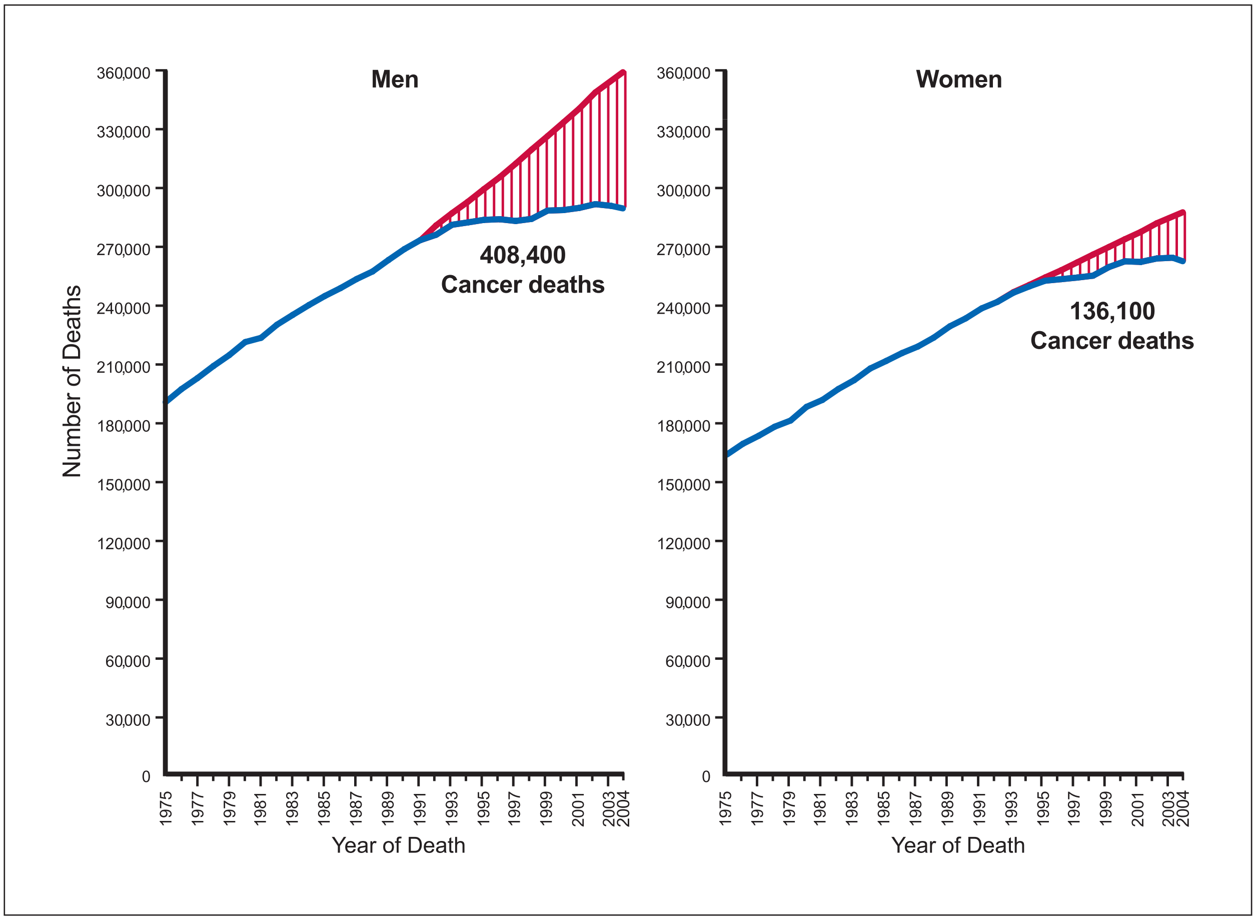


FIGURE **7** Total Number of Cancer Deaths Avoided from 1991 to 2004 in Men and 1992 to 2004 in Women.
The blue line represents the actual number of cancer deaths recorded in each year, and the bold red line represents the expected number of cancer deaths if cancer mortality rates had remained the same since 1990/1991.

Asian American/Pacific Islander women have a greater probability of dying from cancer within 5 years of diagnosis than non-Hispanic Whites, after accounting for differences in stage at diagnosis.[18,24] For the 4 major cancer sites (prostate, female breast, lung and bronchus, and colon and rectum), minority populations are more likely to be diagnosed at distant stage compared with non-Hispanic Whites.[24]

CANCER IN CHILDREN

Cancer is the second most common cause of death among children between the age of 1 and 14 years in the United States, surpassed only by accidents (Table 15). Leukemia (particularly acute lymphocytic leukemia) is the most common cancer in children (aged 0 to 14 years), followed by cancer of the brain and other nervous system, neuroblastoma, renal (Wilms) tumors, and non-Hodgkin lymphoma.[3] Over the past 25 years, there have been significant improvements in the 5-year relative survival rate for all of the major childhood cancers (Table 16). The 5-year relative survival rate among children for all cancer sites combined improved from 58% for patients diagnosed in 1975 to 1977 to 80% for those diagnosed in 1996 to 2003.[3]

LIMITATIONS

Estimates of the expected numbers of new cancer cases and cancer deaths should be interpreted cautiously. These estimates may vary considerably from year to year, particularly for less common cancers and in states with smaller pop-

TABLE **11** Cancer Incidence and Death Rates* by Site, Race, and Ethnicity, United States, 2000 to 2004

	White	African American	Asian American and Pacific Islander	American Indian and Alaska Native†	Hispanic/ Latino‡§
Incidence					
All sites					
Males	556.7	663.7	359.9	321.2	421.3
Females	423.9	396.9	285.8	282.4	314.2
Breast (female)	132.5	118.3	89.0	69.8	89.3
Colon & rectum					
Males	60.4	72.6	49.7	42.1	47.5
Females	44.0	55.0	35.3	39.6	32.9
Kidney & renal pelvis					
Males	18.3	20.4	8.9	18.5	16.5
Females	9.1	9.7	4.3	11.5	9.1
Liver & bile duct					
Males	7.9	12.7	21.3	14.8	14.4
Females	2.9	3.8	7.9	5.5	5.7
Lung & bronchus					
Males	81.0	110.6	55.1	53.7	44.7
Females	54.6	53.7	27.7	36.7	25.2
Prostate	161.4	255.5	96.5	68.2	140.8
Stomach					
Males	10.2	17.5	18.9	16.3	16.0
Females	4.7	9.1	10.8	7.9	9.6
Uterine cervix	8.5	11.4	8.0	6.6	13.8

	White	African American	Asian American and Pacific Islander	American Indian and Alaska Native†	Hispanic/ Latino‡¶
Mortality					
All sites					
Males	234.7	321.8	141.7	187.9	162.2
Females	161.4	189.3	96.7	141.2	106.7
Breast (female)	25.0	33.8	12.6	16.1	16.1
Colon & rectum					
Males	22.9	32.7	15.0	20.6	17.0
Females	15.9	22.9	10.3	14.3	11.1
Kidney & renal pelvis					
Males	6.2	6.1	2.4	9.3	5.4
Females	2.8	2.8	1.1	4.3	2.3
Liver & bile duct					
Males	6.5	10.0	15.5	10.7	10.8
Females	2.8	3.9	6.7	6.4	5.0
Lung & bronchus					
Males	72.6	95.8	38.3	49.6	36.0
Females	42.1	39.8	18.5	32.7	14.6
Prostate	25.6	62.3	11.3	21.5	21.2
Stomach					
Males	5.2	11.9	10.5	9.6	9.1
Females	2.6	5.8	6.2	5.5	5.1
Uterine cervix	2.3	4.9	2.4	4.0	3.3

*Per 100,000 population, age-adjusted to the 2000 US standard population.
†Data based on Contract Health Service Delivery Areas (CHSDA), 624 counties comprising 54% of the US American Indian/Alaska Native population; for more information please see Espey DK, Wu XC, Swan J, et al.[17]
‡Persons of Hispanic/Latino origin may be of any race.
§Incidence data unavailable from the Alaska Native Registry and Kentucky.
¶Mortality data unavailable from Minnesota, New Hampshire, and North Dakota.
Source: Ries LAG, Melbert D, Krapcho M, et al.[3] Incidence data are from the National Cancer Institute; mortality data are from the Centers for Disease Control and Prevention.

TABLE **12** Cancer Death Rates* by Educational Attainment, Race, and Sex, United States, 2001

	Male			Female		
	African American	Non-Hispanic White	Absolute Difference	African American	Non-Hispanic White	Absolute Difference
All sites						
≤ 12 years of education	214.43	163.78	50.65	148.10	128.79	19.31
> 12 years of education	90.14	73.00	17.14	103.30	73.02	30.28
RR (95% CI)	2.38 (2.33-2.43)	2.24 (2.23-2.26)		1.43 (1.41-1.46)	1.76 (1.75-1.78)	
Absolute difference	124.29	90.78		44.80	55.77	
Lung						
≤ 12 years of education	73.23	60.99	12.24	30.82	37.06	-6.24
> 12 years of education	25.78	18.13	7.65	17.92	14.20	3.72
RR (95% CI)	2.84 (2.69-3.00)	3.36 (3.30-3.43)		1.72 (1.61-1.84)	2.6 (2.53-2.67)	
Absolute difference	47.45	42.86		12.90	22.86	
Colorectal						
≤ 12 years of education	20.59	14.23	6.36	14.13	9.36	4.77
> 12 years of education	11.30	7.88	3.42	10.82	5.44	5.38
RR (95% CI)	1.81 (1.63-2.02)	1.81 (1.73-1.89)		1.31 (1.18-1.45)	1.72 (1.63-1.82)	
Absolute difference	9.29	6.35		3.31	3.92	
Prostate						
≤ 12 years of education	10.52	3.26	7.26			
> 12 years of education	4.80	2.22	2.58		NA	
RR (95% CI)	2.17 (1.82-2.58)	1.47 (1.34-1.62)				
Absolute difference	5.72	1.04				
Breast						
≤ 12 years of education				36.06	25.20	10.86
> 12 years of education		NA		31.13	18.50	12.63
RR (95% CI)				1.16 (1.10-1.22)	1.36 (1.32-1.40)	
Absolute difference				4.93	6.70	

Abbreviations: RR, relative risk; CI, confidence interval; NA, not applicable.
*Rates are for individuals aged 25 to 64 years at death, per 100,000, and age-adjusted to the 2000 US standard population.
Source: Adapted from Albano JD, Ward E, Jemal A, et al.[21]

TABLE **13** Probability of Developing Invasive Cancers Within Selected Age Intervals by Sex, United States*

		Birth to 39 (%)	40 to 59 (%)	60 to 69 (%)	70 and Older (%)	Birth to Death (%)
All sites†	Male	1.42 (1 in 70)	8.58 (1 in 12)	16.25 (1 in 6)	38.96 (1 in 3)	44.94 (1 in 2)
	Female	2.04 (1 in 49)	8.97 (1 in 11)	10.36 (1 in 10)	26.31 (1 in 4)	37.52 (1 in 3)
Urinary bladder‡	Male	0.02 (1 in 4,477)	0.41 (1 in 244)	0.96 (1 in 104)	3.50 (1 in 29)	3.70 (1 in 27)
	Female	0.01 (1 in 9,462)	0.13 (1 in 790)	0.26 (1 in 384)	0.99 (1 in 101)	1.17 (1 in 85)
Breast	Female	0.48 (1 in 210)	3.86 (1 in 26)	3.51 (1 in 28)	6.95 (1 in 15)	12.28 (1 in 8)
Colon & rectum	Male	0.08 (1 in 1,329)	0.92 (1 in 109)	1.60 (1 in 63)	4.78 (1 in 21)	5.65 (1 in 18)
	Female	0.07 (1 in 1,394)	0.72 (1 in 138)	1.12 (1 in 89)	4.30 (1 in 23)	5.23 (1 in 19)
Leukemia	Male	0.16 (1 in 624)	0.21 (1 in 468)	0.35 (1 in 288)	1.18 (1 in 85)	1.50 (1 in 67)
	Female	0.12 (1 in 837)	0.14 (1 in 705)	0.20 (1 in 496)	0.76 (1 in 131)	1.06 (1 in 95)
Lung & bronchus	Male	0.03 (1 in 3,357)	1.03 (1 in 97)	2.52 (1 in 40)	6.74 (1 in 15)	7.91 (1 in 13)
	Female	0.03 (1 in 2,964)	0.82 (1 in 121)	1.81 (1 in 55)	4.61 (1 in 22)	6.18 (1 in 16)
Melanoma of the skin	Male	0.15 (1 in 656)	0.61 (1 in 164)	0.66 (1 in 151)	1.56 (1 in 64)	2.42 (1 in 41)
	Female	0.26 (1 in 389)	0.50 (1 in 200)	0.34 (1 in 297)	0.71 (1 in 140)	1.63 (1 in 61)
Non-Hodgkin lymphoma	Male	0.13 (1 in 760)	0.45 (1 in 222)	0.57 (1 in 174)	1.61 (1 in 62)	2.19 (1 in 46)
	Female	0.08 (1 in 1,212)	0.32 (1 in 312)	0.45 (1 in 221)	1.33 (1 in 75)	1.87 (1 in 53)
Prostate	Male	0.01 (1 in 10,553)	2.54 (1 in 39)	6.83 (1 in 15)	13.36 (1 in 7)	16.72 (1 in 6)
Uterine cervix	Female	0.16 (1 in 638)	0.28 (1 in 359)	0.13 (1 in 750)	0.19 (1 in 523)	0.70 (1 in 142)
Uterine corpus	Female	0.06 (1 in 1,569)	0.71 (1 in 142)	0.79 (1 in 126)	1.23 (1 in 81)	2.45 (1 in 41)

*For those individuals free of cancer at beginning of age interval.
†All sites excludes basal and squamous cell skin cancers and in situ cancers except urinary bladder.
‡Includes invasive and in situ cancer cases.
Source: DevCan: Probability of Developing or Dying of Cancer Software, Version 6.2.1. Statistical Research and Applications Branch, National Cancer Institute, 2007. www.srab.cancer.gov/devcan.

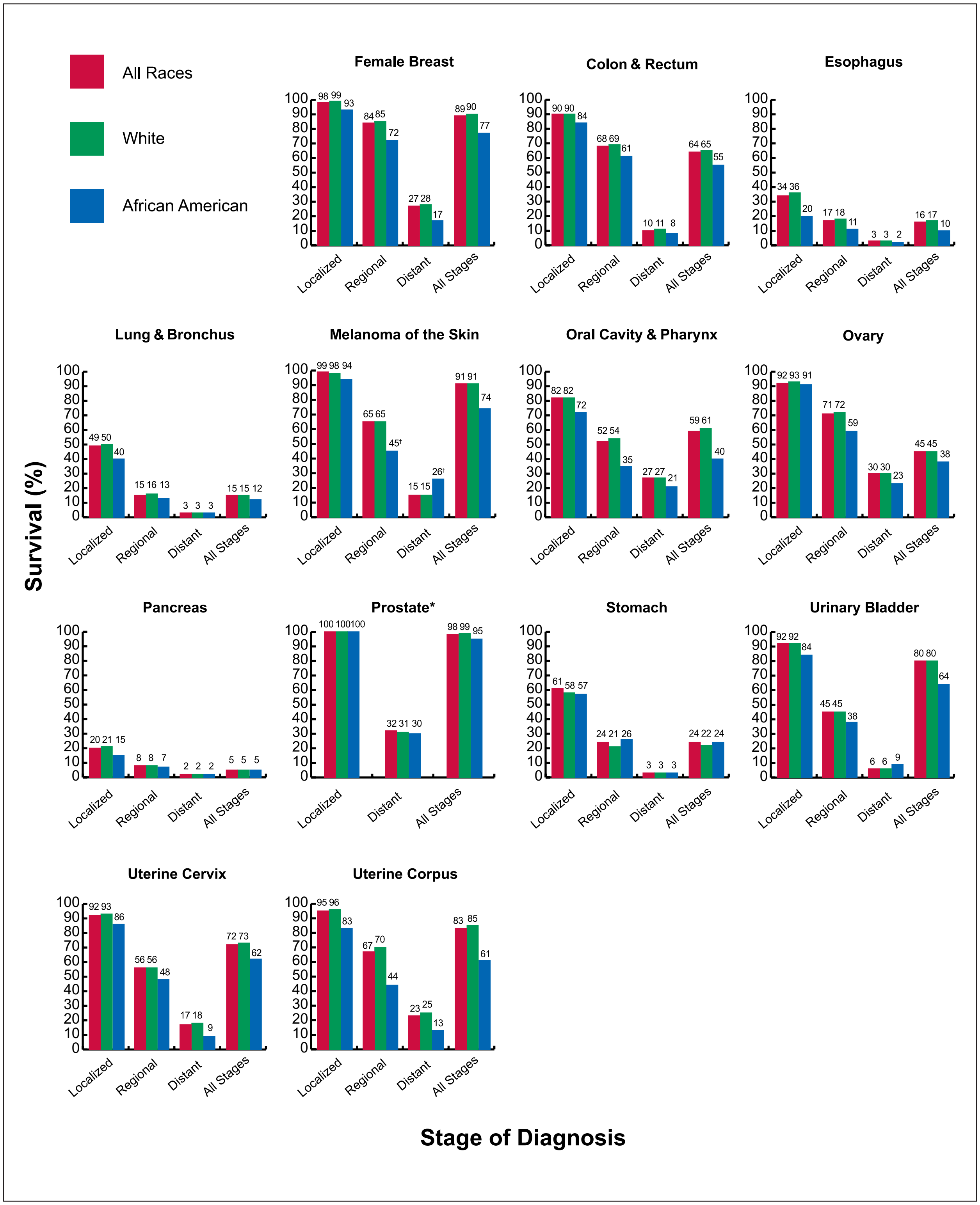


FIGURE **8** Five-year Relative Survival Rates Among Patients Diagnosed with Selected Cancers by Race and Stage at Diagnosis, United States, 1996 to 2003.

* The rate for localized stage represents localized and regional stages combined.

†The standard error of the survival rate is between 5 and 10 percentage points.

Note: Staging according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system. Comparison of this data to that of previous years is discouraged due to the use of an expanded data set.

Source: Ries LAG, Melbert D, Krapcho M, et al.[3]

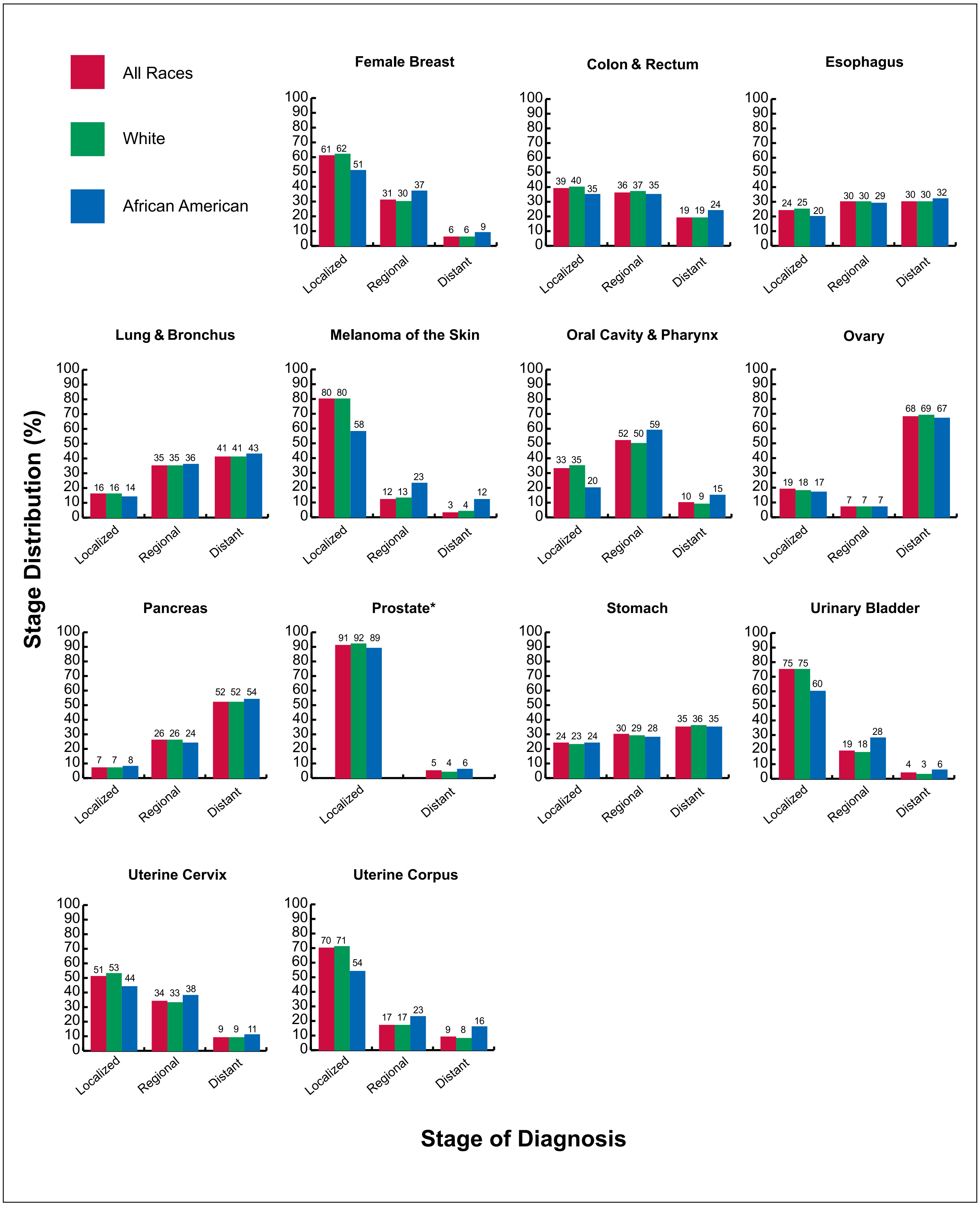


FIGURE **9** Distribution of Selected Cancers by Race and Stage at Diagnosis, United States, 1996 to 2003.
*The rate for localized stage represents localized and regional stages combined.
Note: Staging according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system. For each cancer type, stage categories do not total 100% because sufficient information is not available to assign a stage to all cancer cases. Comparison of this data to that of previous years is discouraged due to the use of an expanded data set.
Source: Ries LAG, Melbert D, Krapcho M, et al.[3]

TABLE **14** Trends in 5-Year Relative Survival Rates* (%) by Race and Year of Diagnosis, United States, 1975 to 2003

	White			African American			All Races		
	1975 to 1977	1984 to 1986	1996 to 2003	1975 to 1977	1984 to 1986	1996 to 2003	1975 to 1977	1984 to 1986	1996 to 2003
All sites	51	55	67 †	40	41	57 †	50	54	66 †
Brain	23	28	34 †	27	33	37 †	24	29	35 †
Breast (female)	76	80	90 †	62	65	78 †	75	79	89 †
Colon	52	60	66 †	46	50	55 †	51	59	65 †
Esophagus	6	11	18 †	3	8	11 †	5	10	16 †
Hodgkin lymphoma	74	80	87 †	71	75	81 †	74	79	86 †
Kidney	51	56	66 †	50	54	66 †	51	56	66 †
Larynx	67	68	66	59	53	50	67	66	64
Leukemia	36	43	51 †	34	34	40	35	42	50 †
Liver & bile duct	4	6	10 †	2	5	7 †	4	6	11 †
Lung & bronchus	13	14	16 †	12	11	13 †	13	13	16 †
Melanoma of the skin	82	87	92 †	60 ‡	70 §	77	82	87	92 †
Myeloma	25	27	34 †	31	32	32	26	29	34 †
Non-Hodgkin lymphoma	48	54	65 †	49	48	56	48	53	64 †
Oral cavity	55	57	62 †	36	36	41	53	55	60 †
Ovary	37	39	45 †	43	41	38	37	40	45 †
Pancreas	3	3	5 †	2	5	5 †	2	3	5 †
Prostate	70	77	99 †	61	66	95 †	69	76	99 †
Rectum	49	58	66 †	45	46	58 †	49	57	66 †
Stomach	15	18	22 †	16	20	24 †	16	18	24 †
Testis	83	93	96 †	82 ‡	87 ‡	88	83	93	96 †
Thyroid	93	94	97 †	91	90	94	93	94	97 †
Urinary bladder	75	79	81 †	51	61	65 †	74	78	81 †
Uterine cervix	71	70	74 †	65	58	66	70	68	73 †
Uterine corpus	89	85	86 †	61	58	61	88	84	84 †

*Survival rates are adjusted for normal life expectancy and are based on cases diagnosed in the SEER 9 areas from 1975 to 1977, 1984 to 1986, and 1996 to 2003, and followed through 2004.
†The difference in rates between 1975 to 1977 and 1996 to 2003 is statistically significant ($P < .05$).
‡The standard error of the survival rate is between 5 and 10 percentage points.
§The standard error of the survival rate is greater than 10 percentage points.
Source: Ries LAG, Melbert D, Krapcho M, et al.[3]

ulations. Estimates are also affected by changes in method. The introduction of a new method for projecting incident cancer cases beginning with the 2007 estimates substantially affected the estimates for a number of cancers, particularly leukemia and female breast (see Pickle et al[12] for more detailed discussion). Not all changes in cancer trends are captured by modeling techniques. For these reasons, we discourage the use of these estimates to track year-to-year changes in cancer occurrence and death. The preferred data sources for tracking cancer trends are the age-standardized or age-specific cancer death rates from the NCHS and cancer incidence rates from SEER or NPCR, even though these data are 3 and 4 years old, respectively, by the time that they become available. Nevertheless, the American Cancer Society estimates of the number of new cancer cases and deaths in the current year provide reasonably accurate estimates of the burden of new cancer cases and deaths in the United States.

TABLE **15** Fifteen Leading Causes of Death Among Children Aged 1 to 14, United States, 2005

Rank	Cause of Death	Number of Deaths	Percent (%) of Total Deaths	Death Rate*
	All causes	**11,358**	**100.0**	**19.9**
1	Accidents (unintentional injuries)	4,079	35.9	7.1
2	Cancer	1,377	12.1	2.4
3	Congenital anomalies	918	8.1	1.6
4	Assault (homicide)	716	6.3	1.2
5	Heart diseases	403	3.5	0.7
6	Intentional self-harm (suicide)	272	2.4	0.5
7	Influenza & pneumonia	216	1.9	0.4
8	Septicemia	166	1.5	0.3
9	Chronic lower respiratory disease	160	1.4	0.3
10	Cerebrovascular disease	157	1.4	0.3
11	In situ & benign neoplasms	128	1.1	0.2
12	Meningitis	58	0.5	0.1
13	Anemias	49	0.4	0.1
14	Diabetes mellitus	40	0.4	0.1
15	Nephritis, nephrotic syndrome, & nephrosis	39	0.3	0.1
	All other causes	2,580	22.7	

*Rates are per 100,000 population and age-adjusted to the 2000 US standard population.
Note: “Symptoms, signs, and abnormal clinical and laboratory findings,” “Events of undetermined intent,” “Certain perinatal conditions,” “Other respiratory diseases,” and “Other and unspecified infectious and parasitic diseases” were excluded from ranking order.
Source: US Mortality Data, 2005, National Center for Health Statistics, Centers for Disease Control and Prevention, 2008.

TABLE **16** Trends in Five-year Relative Survival Rates* (%) for Children Under Age 15, United States, 1975 to 2003

	Year of Diagnosis							
Site	1975 to 1977	1978 to 1980	1981 to 1983	1984 to 1986	1987 to 1989	1990 to 1992	1993 to 1995	1996 to 2003
All sites	58	63	67	68	71	76	77	80 †
Acute lymphocytic leukemia	58	66	71	73	78	83	84	87 †
Acute myeloid leukemia	19	26	27 ‡	31 ‡	37 ‡	41	42 ‡	54 †
Bone & joint	51 ‡	49	57 ‡	59 ‡	67 ‡	67	74	72 †
Brain & other nervous system	57	58	56	62	64	64	70	74 †
Hodgkin lymphoma	81	88	88	91	87	97	95	95 †
Neuroblastoma	52	57	55	52	62	76	67	69 †
Non-Hodgkin lymphoma	43	53	67	70	71	76	81	87 †
Soft tissue	61	75	69	74	65	80	77	72 †
Wilms' tumor	73	79	87	91	92	92	92	92 †

*Survival rates are adjusted for normal life expectancy and are based on follow up of patients through 2004.
†The difference in rates between 1975 to 1977 and 1996 to 2003 is statistically significant ($P < .05$).
‡The standard error of the survival rate is between 5 and 10 percentage points.
Note: “All sites” excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.
Source: Ries LAG, Melbert D, Krapcho M, et al.[3]

REFERENCES

1. National Center for Health Statistics, Division of Vital Statistics, Centers for Disease Control and Prevention. Available at: http://www.cdc.gov/nchs/nvss.htm. Accessed November 15, 2007.

2. National Cancer Institute, US National Institutes of Health. Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Incidence—SEER 9 Regs Public-Use, Nov 2006 Sub (1973–2004), Linked to County Attributes, Total US, 1969–2004 Counties, National Cancer Institute, DCCPS, Surveillance Research Program, Cancer Statistics Branch, released April 2007, based on the November 2006 submission.

3. Ries LAG, Melbert D, Krapcho M, et al., eds. SEER Cancer Statistics Review, 1975–2004, National Cancer Institute. Bethesda, MD, http://seer.cancer.gov/csr/1975_2004/, based on November 2006 SEER data submission, posted to the SEER Web site, 2007.

4. National Cancer Institute, US National Institutes of Health. Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Incidence—SEER 13 Regs Public-Use, Nov 2006 Sub (1992–2004), Linked to County Attributes, Total US, 1969–2004

Counties, National Cancer Institute, DCCPS, Surveillance Research Program, Cancer Statistics Branch, released April 2007, based on the November 2006 submission.

5. National Cancer Institute, US National Institutes of Health. Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov) SEER*Stat Database: Incidence—SEER 17 Regs Public-Use, Nov 2006 Sub (2000–2004), Linked to County Attributes, Total US, 1969–2004 Counties, National Cancer Institute, DCCPS, Surveillance Research Program, Cancer Statistics Branch, released April 2007, based on the November 2006 submission.

6. Wu XC, McLaughlin CC, Lake A, et al., eds. Cancer in North America, 2000–2004. Volume One: Incidence. Springfield, IL: North American Association of Central Cancer Registries, Inc.; 2007.

7. US Census Bureau. Available at: http://www.census.gov. Accessed September 16, 2007.

8. Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death. Vol. 1, 10th revision. Geneva, Switzerland: World Health Organization; 1992.

9. Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death. Vol. 1, 9th revision. Geneva, Switzerland: World Health Organization; 1975.

10. Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death. Vol. 1, 8th revision. Geneva, Switzerland: World Health Organization; 1967.

11. Fritz A, Percy C, Jack A, et al., eds. International Classification of Diseases for Oncology, 3rd ed. Geneva, Switzerland: World Health Organization; 2000.

12. Pickle LW, Hao Y, Jemal A, et al. A new method of estimating United States and state-level cancer incidence counts for the current calendar year. CA Cancer J Clin 2007;57:30–42.

13. Tiwari RC, Ghosh K, Jemal A, et al. A new method of predicting US and state-level cancer mortality counts for the current calendar year. CA Cancer J Clin 2004;54:30–40.

14. Clegg LX, Feuer EJ, Midthune DN, et al. Impact of reporting delay and reporting error on cancer incidence rates and trends. J Natl Cancer Inst 2002;94:1537–1545.

15. Jemal A, Ward E, Thun MJ. Recent trends in breast cancer incidence rates by age and tumor characteristics among U.S. women. Breast Cancer Res 2007;9:R28.

16. Ravdin PM, Cronin KA, Howlader N, et al. The decrease in breast-cancer incidence in 2003 in the United States. N Engl J Med 2007;356:1670–1674.

17. Espey DK, Wu XC, Swan J, et al. Annual report to the nation on the status of cancer, 1975–2004, featuring cancer in American Indians and Alaska Natives. Cancer 2007;110:2119–2152.

18. Jemal A, Clegg LX, Ward E, et al. Annual report to the nation on the status of cancer, 1975–2001, with a special feature regarding survival. Cancer 2004;101:3–27.

19. Ghafoor A, Jemal A, Ward E, et al. Trends in breast cancer by race and ethnicity. CA Cancer J Clin 2003;53:342–355.

20. Ward E, Jemal A, Cokkinides V, et al. Cancer disparities by race/ethnicity and socioeconomic status. CA Cancer J Clin 2004;54:78–93.

21. Albano JD, Ward E, Jemal A, et al. Cancer mortality in the United States by education level and race. J Natl Cancer Inst 2007;99:1384–1394.

22. Ghafoor A, Jemal A, Cokkinides V, et al. Cancer statistics for African Americans. CA Cancer J Clin 2002;52:326–341.

23. Bach PB, Schrag D, Brawley OW, et al. Survival of blacks and whites after a cancer diagnosis. JAMA 2002;287:2106–2113.

24. Clegg LX, Li FP, Hankey BF, et al. Cancer survival among US whites and minorities: a SEER (Surveillance, Epidemiology, and End Results) Program population-based study. Arch Intern Med 2002;162:1985–1993.